

Mapping digital twin applications in infrastructure and the built environment across research types, methods, sectors, phases, and scales

Soheila Kookalani^{a,*}, Stephen Green^a, Peihang Luo^a, Hamidreza Alavi^{a,b}, Erika Parn^a, Zhaojie Sun^a, Ioannis Brilakis^a

^a Department of Engineering, University of Cambridge, Cambridge CB3 0FA, UK

^b School of the Built Environment, Oxford Brookes University, Oxford OX3 0BP, UK

ARTICLE INFO

Keywords:

Digital twin
Built environment
Building information modelling
Artificial intelligence
Infrastructure

ABSTRACT

Digital Twin technologies are increasingly used in infrastructure and the built environment to create dynamic, data-driven models of physical assets and processes. This review analyses recent advancements across sectors such as tunnels, bridges, roads, buildings, construction management, and urban planning, covering all life-cycle phases from design to operation. Integrating Digital Twins with Building Information Modelling, Internet of Things sensors, and Artificial Intelligence enhances real-time monitoring, decision-making, and asset performance. Key methods include monitoring, modelling, and simulation, which improve resource use and proactive maintenance. However, adoption faces challenges such as poor data interoperability, high costs, and technical complexity in merging multiple technologies. Ethical and governance issues around data privacy and security also persist. The review identifies future research needs in improving interoperability, expanding predictive analytics, and assessing large-scale impacts. It highlights Digital Twins' potential to improve resilience, efficiency, and sustainability, stressing the need for policy support and stakeholder collaboration.

1. Introduction

The emergence of Digital Twin (DT) technologies represents a transformative shift in the infrastructure and built environment sectors. Originally conceptualized by Grieves in 2003 as a three-dimensional model linking physical and virtual assets [1], DTs have evolved into complex, data-driven systems integrating the Internet of Things (IoT), Artificial Intelligence (AI), and big data technologies [2–4]. This progression has enabled DTs to transcend static representations, becoming dynamic, real-time platforms for simulation, optimization, and decision support across the entire asset lifecycle, from design and construction to operation, maintenance, and eventual decommissioning [5–8]. Their increasing adoption across sectors underscores DTs' growing role in modernizing infrastructure systems and advancing sustainable, resilient built environments.

DTs have gained considerable traction in high-stakes domains such as tunnels [9,10], bridges [11,12], and buildings [13,14] owing to their ability to integrate real-time monitoring, simulation, and decision support. For example, semantic modelling and rule-based reasoning have been used to enhance tunnel operations [15], while vibration-based

health monitoring has enabled rapid post-earthquake assessments. These applications demonstrate strength of DTs in supporting context-aware, data-rich decision-making in critical environments and highlight their potential to improve operational reliability, safety, and performance in complex infrastructure systems.

In parallel, the civil engineering and built environment communities have witnessed a growing interest in the integration of AI into DT frameworks [16–20]. AI empowers DTs with predictive and prescriptive analytics, facilitating real-time anomaly detection, fault diagnosis, and autonomous decision-making [21–25]. From optimizing energy performance and material usage to improving scheduling and risk management, AI acts as a cornerstone of the next generation of intelligent, adaptive infrastructure. The intersection of AI and DTs thus represents a fertile area for innovation, where machine learning (ML) and knowledge-driven reasoning unlock higher levels of automation, responsiveness, and resilience.

The integration of DTs with Building Information Modelling (BIM), IoT, and AI has substantially expanded their functional and operational scope [26–31]. BIM provides a structured, semantically rich foundation for representing physical assets; IoT enables continuous data acquisition

* Corresponding author.

E-mail address: sk2268@cam.ac.uk (S. Kookalani).

<https://doi.org/10.1016/j.autcon.2026.106778>

Received 28 July 2025; Received in revised form 7 January 2026; Accepted 9 January 2026

Available online 13 January 2026

0926-5805/© 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

and synchronization; and AI techniques drive predictive maintenance, performance optimization, and automated control [32,33]. This technological synergy enhances situational awareness, supports proactive interventions, and enables adaptive design across diverse contexts, from energy management in smart buildings to safety-critical infrastructure monitoring [34].

Despite these advancements, several barriers continue to hinder widespread DT implementation in infrastructure contexts. The absence of standardized frameworks for data interoperability, the complexity of integrating multidisciplinary technologies, and the high costs associated with large-scale deployment remain critical challenges [35]. Quantitative maturity models, particularly within underground and transport infrastructure, have been proposed to assess readiness and identify capability gaps, yet these frameworks remain limited in scope [36]. Addressing such systemic issues is essential to transition DTs from isolated pilots toward scalable, institutionalized systems capable of supporting cross-sector digital transformation.

Emerging technological paradigms such as federated learning, edge computing, and hybrid AI pipelines offer promising pathways to overcome these challenges [37–39]. These approaches can reduce computational demand, enhance data privacy, and enable more adaptive and decentralized DT architectures. Developing lightweight, scalable, and interoperable frameworks will democratize DT adoption, particularly within resource-constrained infrastructure systems. As these innovations mature, coordinated multi-stakeholder collaboration will be essential to ensure technical robustness, policy alignment, and long-term sustainability.

Several recent reviews have discussed DT applications in the built environment, such as AlBalkhy et al. [35] on DTs in the built environment, Wang et al. [40] on DTs for road infrastructure, and Liu et al. [41] on DTs in urban management. However, these studies typically focus on single sectors or specific technical themes [42,43]. In contrast, the

present review adopts a cross-sectoral perspective encompassing tunnels, bridges, roads, construction management, urban planning, and buildings. It systematically maps DT research across research types, methods, lifecycle phases, and spatial scales, providing a multi-dimensional synthesis that is currently absent in existing reviews. This approach allows for identifying knowledge gaps and methodological trends across domains, offering a comprehensive reference framework for future DT research and standardization efforts. Given this context, the present review aims to synthesize and classify the state of the art in DT research within the infrastructure and built environment domains. While interest in DTs has surged in recent years, the literature remains fragmented in definitions, implementation approaches, and evaluation criteria.

To clarify this landscape, the study addresses the following research questions:

- **RQ1:** How have DTs been applied to infrastructure and the built environment since 2020?
- **RQ2:** What methods, tools, and data pipelines support these DTs?
- **RQ3:** What challenges impede broader adoption and where should future research focus?

To address these questions, the paper is organized as follows: [Section 2](#) outlines the research methodology used to identify and analyse relevant literature. [Section 3](#) categorizes existing studies into theoretical, conceptual, technical, and literature review papers. [Section 4](#) discusses the core methods, including monitoring, modelling, and simulation, underpinning DT implementations. [Section 5](#) examines sector-specific applications, covering tunnels, bridges, roads, construction management, urban planning, and buildings. [Section 6](#) explores the lifecycle phases of DT deployment, while [Section 7](#) evaluates the scalability of DTs across object, asset, and city levels. Finally, [Section 8](#) presents a

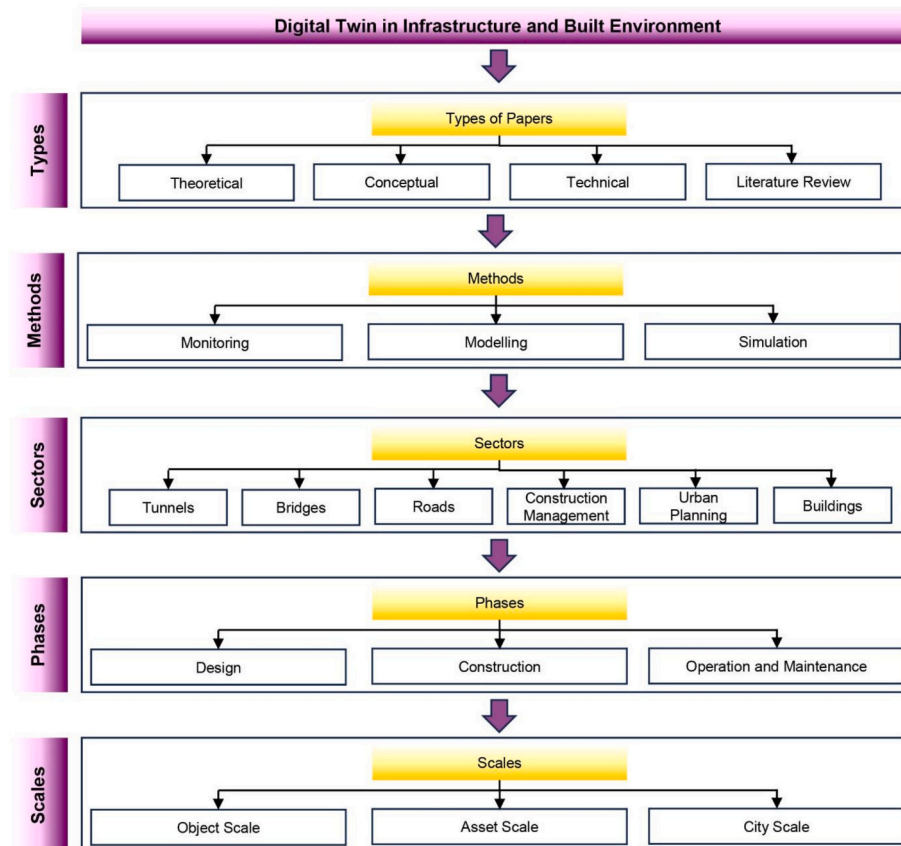


Fig. 1. Overall framework of this research.

discussion of current challenges and future research directions, and [Section 9](#) concludes the paper by summarizing the key insights. [Fig. 1](#) presents the overall framework of this study. Together, these sections aim to offer a comprehensive, multi-dimensional review that advances understanding and promotes coherent development of DT applications in the infrastructure and built environment.

2. Research methodology

This section outlines the systematic methodology employed to conduct a comprehensive literature review on the application of DT technologies in the infrastructure and built environment fields. The methodology ensures the selection of relevant and high-quality research studies by adhering to specific criteria and a structured search process. The review focuses on the design, construction, operation, and maintenance of infrastructure and the built environment, with emphasis on tunnels, bridges, roads, buildings, construction management, and urban planning. Key areas of interest include monitoring, modelling, and simulation techniques within these contexts.

2.1. Database selection

To ensure a comprehensive review, the study utilized various academic databases, including Scopus, Web of Science, and Google Scholar. Among these, the Web of Science Core Collection was prioritized for its multidisciplinary coverage and reliable citation indexing, which encompasses journals, books, and conference proceedings of high scholarly standards. Searches conducted within the Web of Science Core Collection formed the backbone of the review, ensuring access to contemporary, peer-reviewed literature on DT.

2.2. Search process

The search strategy began with a targeted exploration of DT applications in construction management. The initial database query combined the keywords “Digital Twin” AND “construction management” in the Web of Science Core Collection. To ensure comprehensiveness, the following Boolean combinations were subsequently used: “Digital Twin” AND (“infrastructure” OR “built environment”), “Digital Twin” AND (“bridge” OR “tunnel” OR “road”), “Digital Twin” AND (“urban planning” OR “smart city”), “Digital Twin” AND (“BIM” OR “Building Information Modelling”), “Digital Twin” AND (“operation and maintenance” OR “monitoring” OR “simulation”). Each search string was applied to **Title, Abstract, and Keywords**, filtered for English-language publications between **January 2020 and June 2024**. The combined searches across Web of Science, Scopus, and Google Scholar resulted in **341 initial records**. To mitigate potential disciplinary bias introduced by construction-oriented terms, **backward and forward citation searches** were performed.

- **Backward search:** reference lists of the seed papers were manually examined, following up to **two levels** of cited references.
- **Forward search:** subsequent papers citing the seed papers (using Web of Science “Cited Reference Search”) were traced for up to **one level**. The process was discontinued when additional rounds of reference tracing yielded only duplicates or papers outside the DT domain.

These systematic backward–forward searches expanded the dataset to **104 relevant studies** after applying inclusion and exclusion criteria. To verify that no substantial thematic shifts occurred after June 2024, automatic database alerts and forward-citation checks were conducted up to August 2025, with key subsequent publications discussed in the text. These confirmed that later studies primarily reinforce trends already identified, such as AI-driven analytics, BIM-IoT integration, and lifecycle interoperability. All records were managed using **Mendeley**, with duplicate removal and coding conducted manually by two

reviewers to ensure consistency. This multi-stage strategy, combining keyword-based database searches with structured citation chaining, ensured reproducibility, and minimized disciplinary bias while maintaining transparency of selection boundaries.

2.3. Inclusion and exclusion criteria

A rigorous set of inclusion and exclusion criteria was developed to ensure relevance and quality. The inclusion criteria consist of peer-reviewed journal articles and conference papers, studies explicitly addressing DTs in the infrastructure and built environment sectors, and papers published between January 2020 and June 2024, highlighting recent advancements in DT. The exclusion criteria include publications unrelated to infrastructure or built environment applications, studies lacking a focus on monitoring, modelling, or simulation methods, and articles in languages other than English.

In this review, a DT is defined as a dynamic, data-driven digital representation of a physical asset, system, or process that maintains bidirectional connectivity throughout the asset lifecycle to enable monitoring, simulation, and decision support. This definition distinguishes DTs from related but narrower concepts such as BIM, which focuses on static geometric and semantic representation, and IoT systems, which emphasize sensing without integrated simulation or lifecycle feedback. Accordingly, only studies that explicitly demonstrated data-driven coupling between a physical and digital system, via monitoring, modelling, or simulation, were included. Papers limited to BIM modelling without feedback loops or to isolated sensor networks without virtual representation were excluded. This screening process resulted in a final selection of 104 papers, capturing the surge of research activity in recent years and providing insights into both contemporary and earlier developments.

2.4. Screening strategy

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework guided the screening process, as outlined in [Fig. 2](#). This approach involved: Identification of relevant studies through database searches; Screening for duplicates and irrelevant papers; Assessing eligibility based on inclusion and exclusion criteria; Finalizing the papers for review.

To ensure consistency and minimize bias during the screening process, two reviewers independently conducted title/abstract and full-text assessments. Disagreements at either stage were discussed and resolved through consensus. In cases where consensus could not be reached, a third reviewer was consulted to make the final decision. Inter-reviewer consistency was periodically checked by cross verifying a random subset of studies to confirm the reliability of inclusion judgments.

2.5. Keyword analysis

The literature search relied on a combination of keywords, including “digital twin” and “infrastructure” or “built environment”. A network map of keywords was generated using VOSviewer, as shown in [Fig. 3](#). Nodes in the network represent frequently used keywords, with larger nodes indicating higher occurrences. Links between nodes illustrate co-occurrence frequencies, highlighting areas of concentrated research interest. This analysis revealed key research clusters, offering insights into the primary focus areas of DT applications within the infrastructure and built environment domains.

2.6. Analytical framework and classification criteria

Explicit criteria were established for each analytical dimension to ensure methodological transparency and internal consistency in categorizing the reviewed literature. Each paper was evaluated according to its dominant focus, methodological approach, sectoral context, lifecycle

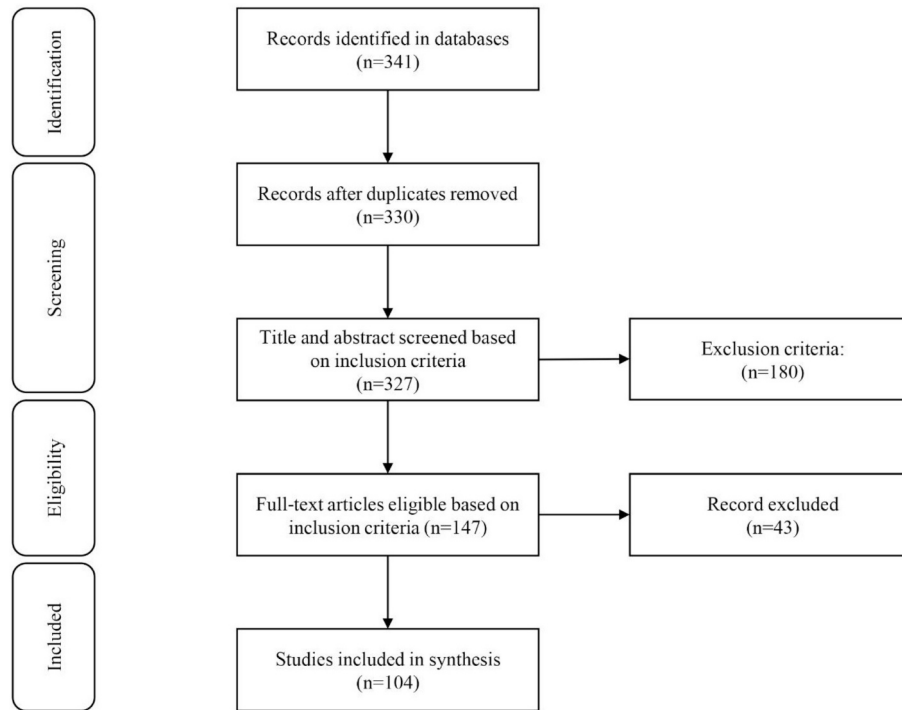


Fig. 2. Screening strategy based on the searching protocol.

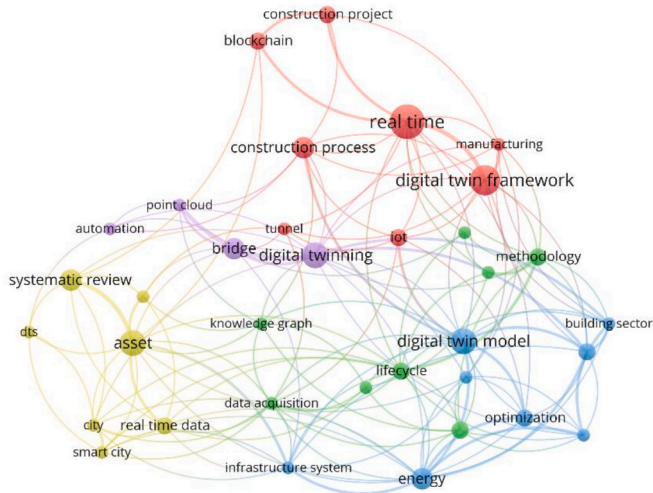


Fig. 3. Main research interest clusters using co-occurrence network of keywords.

phase, and spatial scale. The classification process was guided by the following definitions and boundaries:

2.6.1. Research types

Papers were assigned based on their primary contribution. Theoretical papers propose frameworks, ontologies, or conceptual models without empirical implementation. Conceptual papers present position statements, future outlooks, or synthesis of adoption drivers and challenges. Technical papers describe implemented systems, prototypes, or case-based applications demonstrating DT integration. Literature review papers systematically or narratively synthesize existing studies and identify knowledge gaps.

2.6.2. Methods

Classification was based on the methodological approach to DT development. Monitoring studies focus on real-time data acquisition through sensors or IoT systems. Modelling refers to BIM- or ontology-based digital representations used for lifecycle integration. Simulation includes physics-based or AI-driven scenario analyses and performance forecasting. Papers employing multiple techniques were cross classified to capture methodological overlap.

2.6.3. Sectors

Each study was mapped to its principal application domain, including tunnels, bridges, roads, construction management, urban planning, or buildings. When a paper addressed several domains, it was classified under the dominant sector or marked as multi-sector.

2.6.4. Phases

Papers were categorized according to the infrastructure lifecycle stage addressed, including design, construction, or operation and maintenance. Studies spanning multiple stages were included in all relevant categories.

2.6.5. Scales

The scale dimension reflects the spatial and organizational scope of the DT implementation. Object-scale refers to component-level systems. Asset-scale denotes entire infrastructure entities such as bridges or buildings. City-scale encompasses interconnected urban systems or multi-asset networks.

This multidimensional structure therefore provides a comprehensive mapping framework capable of linking research intent, methodological depth, sectoral application, lifecycle relevance, and implementation scale. It was preferred over purpose-oriented classifications to maintain alignment with the technical, lifecycle-integrated nature of DT research and to facilitate cross-comparison across sectors and phases where common operational taxonomies may not apply.

It should be noted that several papers were classified under more than one category within a given analytical dimension. For instance, a single paper may include both conceptual and technical elements, or

address multiple lifecycle phases (e.g., design and operation). Therefore, the sum of category counts may exceed the total of 104 papers. This intentional overlap reflects the interdisciplinary and multi-phase nature of many DT studies rather than a miscount.

The analytical dimensions described above were explicitly aligned with the three research questions to enhance transparency and replicability. RQ1 was addressed through the classification of studies by sector, phase, and scale, capturing application contexts and lifecycle distribution. RQ2 was analysed through the methods and research type dimensions, examining how monitoring, modelling, and simulation approaches underpin different implementations. RQ3 was derived from cross-comparative synthesis across all dimensions, supported by the discussion sections of each analytical theme (Sections 4–8), where barriers, gaps, and future directions were extracted. This structured linkage ensures that the analytical framework systematically responds to each research question and allows full reproducibility of the review process.

3. Types of papers

The literature in the field of DT and construction technologies presents a diverse body of work, spanning theoretical frameworks, technical solutions, and comprehensive conceptual analyses. This review aims to categorize and summarize existing studies to provide an overview of the current landscape, highlighting the different types of contributions that have been made by researchers in this area. Fig. 4 provides insight into the nature and focus of academic publications. It shows that most papers are technical, with fewer focusing solely on theory, literature review, or conceptual contributions. The overlaps reveal multidisciplinary or hybrid papers that combine elements like theory and literature review or concept and technical application. Table 1 presents a typological overview of the reviewed literature to structure the diverse body of DT-related research. Most studies fall into the technical category, reflecting the field's current emphasis on practical applications and empirical validation. In contrast, theoretical and conceptual studies, though fewer in number, provide the foundational scaffolding needed to scale and generalize DT implementations. Literature review papers play a bridging role, synthesizing fragmented knowledge across disciplines. This distribution suggests a field rich in engineering solutions but still maturing in conceptual cohesion and long-term standardization.

The analysis revealed a diverse distribution of paper types across the reviewed literature. In total, 7 papers were classified as theoretical, 17 as conceptual, 88 as technical, and 15 as literature review papers. Among these categories, technical papers were the most prevalent, reflecting a strong emphasis on applied and implementation-oriented research within the field. Theoretical papers were predominantly published in

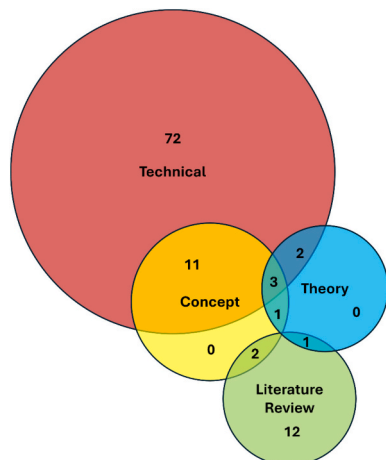


Fig. 4. Classification of the reviewed DT research papers by type.

Table 1

Classification and characteristics of reviewed DT research papers.

Type	Focus Area	Typical Contributions	Common Journals	Ref
Theoretical	Framework development, ontologies, conceptual models	Formal models, DT architecture proposals, adoption frameworks	<i>Buildings, Automation in Construction, Sustainable Cities and Society</i>	[14,44–47]
Conceptual	Adoption drivers, future outlook, interdisciplinary synthesis	Positioning papers, roadmaps, SLR-based reflections	<i>Buildings, Smart Cities, Journal of Building Engineering</i>	[48–50]
Technical	Implementation of DT systems, case studies, integration	AI-driven monitoring, BIM-IoT integration, predictive maintenance systems	<i>Automation in Construction, IEEE Access, Journal of Building Simulation</i>	[51–54]
Literature Review	Domain-specific reviews and state-of-the-art summaries	Systematic literature reviews, classification frameworks, identification of research gaps	<i>Automation in Construction, Journal of Construction Engineering and Management</i>	[55–58]

Buildings, *Journal of Construction Engineering and Management*, *Automation in Construction*, and *Sustainable Cities and Society*, while conceptual studies frequently appeared in *Automation in Construction*, *Buildings*, and the *Journal of Building Engineering*. Technical papers were most found in *Automation in Construction*, *IEEE Access*, and the *Journal of Building Performance Simulation*, showcasing the sector's technological focus. Literature review papers, on the other hand, were mainly published in *Automation in Construction*, *Journal of Construction Engineering and Management*, and *Buildings*, highlighting these journals as key platforms for synthesizing knowledge and advancing theoretical and methodological discussions.

3.1. Theoretical

The theoretical foundation of DTs is rooted in cyber-physical systems, where physical assets and their virtual counterparts interact through continuous, bi-directional data exchange. This connection enables real-time synchronization, monitoring, and intelligent decision-making across the asset lifecycle. Modern DT theory extends beyond representation to emphasize autonomy, learning, and system resilience. Gao et al. [11] introduced an AIoT-informed communication model highlighting information hierarchy and latency reduction via edge computing. Honghong et al. [47] conceptualized a DT-enhanced BIM framework integrating design, construction, and operation, while Esmaeili and Simeone [46] defined Construction DTs as active control systems for predictive management. Zhang et al. [45] advanced analytical capability of DTs through reinforcement learning-based damage detection. Agrawal et al. [50] expanded the theory with a human-centred model, defining roles of observation, analysis, decision, and action. Complementary studies by Lian and Yan [44] and Daniotti et al. [14] reinforced theoretical bases of DTs in education, interoperability, and Lean Construction 4.0. Collectively, these works establish DTs as intelligent, adaptive, and socio-technical systems that unify cybernetics, AI, and human interaction to transform infrastructure management across its full lifecycle.

3.2. Conceptual

The analysis of conceptual contributions identified 17 papers that focus on theoretical discussions and future trajectories of DT applications across the built-environment domain [11,14,48,49,57,59–66]. Collectively, these works frame DTs as socio-technical systems rather than purely technical artefacts, emphasizing governance, interoperability, lifecycle integration, and value realisation at asset and city scales. Position pieces and outlook articles articulate organizational readiness and capability gaps, propose classification and maturity perspectives, and outline integration pathways between BIM/IFC, IoT sensing, and analytics, aimed at moving from pilots to scalable deployment [11,14,60–63]. Several contributions take a city- or programme-level stance, discussing urban DT visions, multi-asset coordination, and implications for resilience and sustainability [57,64,65]. Others focus on sectoral or lifecycle themes such as operations/facility management, safety and performance monitoring, and strategic decision support, offering high-level architectures and research agendas to guide implementation [48,49,59,66].

Several works articulated how DTs should be framed and assembled, often as design-time or governance frameworks rather than operational deployments. For instance, Alnaser et al. [59] outlined a BIM-to-DT direction of travel, while Daniotti et al. [14] took the inverse route by using BIM ontologies to structure a DT for construction renovation workflows. Zhou et al. [62] extended the conversation to urban settings by showing how social attributes could be simulated and integrated into DT research agendas.

Together, these conceptual studies provide the scaffolding for subsequent technical work: they clarify terminology, foreground data and process standards, and prioritise research on interoperability, ethics, and evidence of impact, thereby charting a coherent agenda for DT adoption across design, construction, and operations.

3.3. Technical

The analysis of technical contributions identified 88 papers categorized into several primary themes. The largest category, AI and emerging technologies, comprises papers exploring the integration of cutting-edge technologies such as AI, ML, blockchain, and IoT in DT applications [15,53,54,67–75]. These studies highlight the transformative potential of AI-driven decision support, predictive monitoring, and smart infrastructure systems. The category building performance and energy management includes a paper focused on enhancing energy efficiency and sustainability in urban environments [76]. Construction process and operation efficiency features several papers that delve into improving decision support systems and operational effectiveness in construction [77,78]. The data integration and interoperability category, also comprising a paper, addresses challenges related to seamless data exchange across DT platforms [79]. Monitoring, maintenance, and assessment is represented by 1 theoretical paper focused on predictive maintenance and occupant comfort using DTs [80]. Two papers fall under urban infrastructure and smart cities, emphasizing the application of DTs to urban systems and infrastructure development [80,81]. BIM and data integration consists of several papers that delve into the integration of BIM with DT frameworks for improved data interoperability and management [51–54,58]. Monitoring and predictive maintenance includes 5 papers that demonstrate real-time monitoring systems and predictive maintenance frameworks using DT technologies [70,71,73]. Lastly, the other category consists of several papers that provide unique insights into specialized applications, such as visualization techniques, fault detection, and system-specific diagnostics, further expanding the technical contributions to the field of DTs [51,52,58]. Collectively, these technical papers highlight the significant role of DTs in facilitating technological innovation, operational improvements, and advanced data integration across built environments.

3.4. Literature review

The analysis of literature review papers identified 15 contributions explicitly categorized as systematic reviews or literature reviews. The largest category, construction and infrastructure, comprises papers that provide systematic reviews of DT adoption drivers, operational efficiencies, and readiness frameworks within the architecture, engineering, and construction (AEC) sector [55]. Building performance and sustainability includes a paper that reviews how DTs contribute to improving energy efficiency and environmental monitoring. The maintenance and fault detection category features a paper focusing on predictive maintenance frameworks for building operations [56]. Finally, urban infrastructure and smart cities is represented by a paper that explores the challenges and potential of implementing DTs within urban systems [57]. Collectively, these literature reviews offer valuable insights into current research trends, technological challenges, and future research directions for DT applications across various domains.

3.5. Discussion

The broader typological landscape that emerges from Section 3 illustrates a research field propelled more by practical problem-solving than by theoretical consolidation. Roughly three-quarters of the corpus consists of technical studies centred on applied themes, AI-assisted decision support, BIM-enabled data integration, real-time monitoring infrastructures, and predictive maintenance. This prevalence reflects growing confidence that DT concepts are sufficiently mature for real-world deployment and stress-testing on operational assets. However, it also highlights an enduring limitation: many technical solutions remain highly domain-specific and rely on bespoke data architectures, impeding their transferability across sectors or asset classes.

By contrast, only 7 theoretical and 17 conceptual papers were identified. Although modest in number, these contributions provide essential scaffolding, including reference architectures, ontologies, capability models, and adoption-readiness frameworks, that enable generalisation of insights beyond individual case studies. Their scarcity underscores a clear need for more foundational work to harmonize definitions, establish standardized metrics, and formalize evaluation protocols. Meanwhile, the 15 literature review papers function as an intermediary layer, synthesizing adoption drivers and cross-disciplinary challenges spanning construction, infrastructure, and emerging urban-scale applications.

Recent publications confirm that the typological patterns identified in this section have continued along the same trajectory. Theoretical studies persist in refining foundational DT constructs without departing from their central themes, while conceptual papers sustain their focus on governance frameworks, interoperability requirements, and lifecycle integration challenges [82–84]. Technical papers remain overwhelmingly dominant, expanding work on AI-enabled monitoring pipelines, BIM-IoT integration, and predictive-maintenance workflows [85–89]. At the same time, emerging literature reviews primarily consolidate sector-specific challenges rather than introducing new conceptual directions [40–43,82,90]. Taken together, these developments reinforce the bottom-heavy typological distribution observed in this review, indicating that the relative proportions of theoretical, conceptual, technical, and review contributions have remained broadly consistent.

Taken collectively, the distribution suggests that current DT scholarship is distinctly “bottom-heavy”: abundant in engineering ingenuity but still consolidating the theoretical and conceptual coherence required for systematic knowledge accumulation. A more mature research ecosystem will require stronger bidirectional feedback loops, where empirical insights from technical implementations inform the development of generalisable theory, and theoretically grounded models, in turn, guide the design of interoperable, scalable, and ethically robust DT solutions. Closing this gap will be essential for advancing DTs from

isolated, domain-tuned exemplars toward a cohesive, transferable, and lifecycle-integrated digital infrastructure paradigm.

4. Methods

The literature reveals that various methods can be used to construct a DT in infrastructure and built environment. In these methods, monitoring, modelling, and simulation techniques were commonly used. According to all the investigated papers, the Venn diagram showing the number of papers using different methods to create a DT was shown in Fig. 5. It can be concluded that the monitoring method was the most used, followed by the modelling method and the simulation method. Furthermore, a considerable number of papers combined monitoring and modelling methods for DT development, while few papers combined monitoring and simulation methods. In addition, several papers used all the three methods for the creation of a DT. Table 2 summarizes and contrasts the monitoring, modelling, and simulation methods to synthesize the core findings on methodological approaches in DT development. Each offers distinct advantages and trade-offs depending on the use case. While monitoring excels in capturing real-time conditions, modelling ensures semantic consistency, and simulation supports predictive foresight. Increasingly, advanced DT implementations adopt hybrid workflows to combine these strengths, though integration challenges remain a key bottleneck to scalable deployment.

4.1. Monitoring

Building a DT for infrastructure or buildings hinges critically on the integration and utilization of monitoring techniques and real-time data acquisition. A DT is essentially a virtual representation of a physical asset that maintains real-time synchronization between the digital and physical environments. This synchronization is made possible by employing advanced monitoring techniques and data transmission technologies, such as IoT sensors and wireless communication [51,97].

The integration of monitoring data is at the core of the DT's functionality. Embedded sensors within the physical structure are deployed to collect essential real-time data, including parameters like displacement, strain, and temperature. The data collected by sensors will be input into the DT platform, which ensures that the digital representation remains continuously updated and reflective of the actual state of the physical asset [12,92]. The seamless flow of monitoring data into the DT creates a dynamic and responsive model capable of adapting to real-time changes and conditions [15,51].

The technological framework supporting a DT emphasizes monitoring techniques by starting with BIM as the foundational structure that maps out the physical asset's geometry and attributes. Enhanced data description capabilities, such as Sensor Model Language (SensorML), can be integrated to facilitate detailed and real-time environmental

sensing [15,51,60]. This collected sensor data is then analysed using ML and predictive analytics, underlining the critical role of monitoring in detecting anomalies and forecasting potential failures [80,92,98].

For the structural health evaluation of infrastructure, the role of monitoring is further highlighted by the use of Finite Element Models (FEM). The FEM combined with real-time sensor data enable the DT to simulate the asset's behaviour under different loading conditions and extend analysis to unmonitored sections [92]. For road projects, DTs leverage monitoring data to conduct activities like clearance checking during highway expansions. This data-driven approach significantly reduces the need for extensive field surveys and expedites the design process by using remote sensing and existing data [97].

Monitoring data also plays a pivotal role in the predictive maintenance capabilities of DTs. DTs can anticipate structural issues and schedule maintenance proactively by integrating historical data from sensors and applying predictive algorithms [67,80,91]. This data-driven maintenance strategy not only enhances operational efficiency, but also ensures safety and longevity by keeping the digital model tightly aligned with the physical state of the structure.

In summary, monitoring techniques and data are indispensable to the functioning of a DT, serving as the lifeblood that keeps the digital model up to date. They enable continuous monitoring, predictive analysis, and proactive management of infrastructure, ensuring that the digital representation reflects real-time conditions and supports informed decision-making for safety and efficiency.

4.2. Modelling

Modelling techniques, such as BIM, are commonly used to create a dynamic and accurate virtual representation of physical assets. The central role of modelling techniques is fundamental to the process, providing the foundation for the DT and enabling comprehensive lifecycle management. BIM serves as the core structure for DT creation by offering semantically rich, data-intensive models that capture the geometric, spatial, and functional details of physical assets. As a parametric modelling approach, BIM allows for the integration of design, construction, and operation data into a unified platform, facilitating efficient management across an asset's lifecycle. An open and neutral data schema called Industry Foundation Classes (IFC) is widely adopted to ensure the interoperability and detailed representation of BIM data, enabling seamless data sharing and updates among stakeholders and systems [79,99].

A crucial use of BIM in DT creation is the generation of as-is models of existing structures. In this process, techniques such as integrating CAD drawings with image-based methods, including Structure from Motion (SfM) and Simultaneous Localization and Mapping (SLAM), can be used to enhance geometric accuracy and streamline model creation. These techniques can be leveraged by semi-automatic methods to extract structural information from documents, CAD files, and 3D laser scans, converting them into BIM-compatible formats. This approach facilitates efficient modelling of existing conditions while minimizing manual efforts and errors [48,97]. BIM's capabilities can be further enriched by semantic integration, such as the combination with the Sensor Model Language (SensorML). This enhancement enables the incorporation of real-time data from IoT sensors and monitoring devices, transforming static models into dynamic DTs. These enriched models are invaluable for decision-making in facility management and infrastructure operation, particularly in scenarios requiring real-time environmental monitoring and predictive maintenance [15,48].

Modelling techniques like BIM also support specific applications of DTs across various domains. For instance, BIM-based DTs have been utilized for clearance checking in highway widening projects. This eliminates the need for expensive field surveys by integrating data from online maps, aerial photographs, and digital terrain models. Such applications streamline design processes and enhance decision-making in complex infrastructure projects [97]. Furthermore, automation can

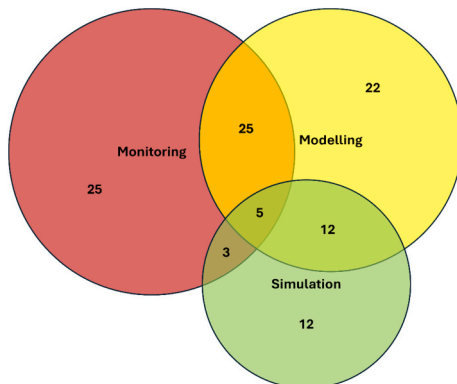


Fig. 5. Venn diagram of number of papers using different methods to create a DT.

Table 2
Comparative summary of DT development methods.

Method	Purpose	Key Technologies	Strengths	Limitations	Ref
Monitoring	Real-time data acquisition and condition assessment	IoT sensors, wireless networks	Enables real-time synchronization; supports predictive maintenance	High sensor deployment cost; issues with data latency and standardization	[91–93]
Modelling	Virtual representation of physical assets and environments	BIM, IFC, semantic ontologies	Semantically rich digital models; supports lifecycle integration	Requires accurate input data; interoperability challenges with legacy systems	[48,53,54,79]
Simulation	Scenario analysis and performance forecasting	FEM, AI/ML, surrogate models	Supports predictive analytics and decision-making in extreme or future states	Computationally intensive; depends on reliable calibration and input data	[62,94]
Hybrid	Combined method leveraging the strengths of all three	Integrated pipelines (BIM + IoT + AI)	Holistic view; dynamic updates and decision support; supports automation	Integration complexity; need for lightweight interoperability frameworks	[50,95,96]

significantly strengthen BIM's role in DT creation, particularly for large-scale or complex environments. Automated and semi-automated methods, such as image-based point cloud processing and object recognition, allow for the rapid and accurate generation of BIM models. These innovations reduce the time and effort required for manual modelling, improving precision and making BIM a scalable solution for DT development [48,79]. In addition, BIM-integrated DTs play a crucial role in visualization and decision support. They offer interactive platforms for stakeholders to monitor, analyse, and manage asset performance effectively. Advanced visualization tools within BIM models enable applications such as HVAC optimization, occupant comfort analysis, and infrastructure safety assessments, providing valuable insights for proactive management [80,97].

In conclusion, the use of modelling techniques like BIM is indispensable in the creation and operation of DTs. BIM forms the backbone of DT technology by integrating advanced data processing, real-time monitoring, semantic enrichment, and automation techniques. It drives precision, proactive management, and innovation across the asset lifecycle, enhancing efficiency and sustainability in the built environment.

4.3. Simulation

DTs also rely on simulation to have insights on the real condition of physical assets. Simulations, particularly those based on the FEM, can replicate the behaviours of physical assets under various operational conditions. This capability allows DTs to serve as powerful analytical tools, simulating real-world scenarios with high accuracy. For example, FEM can be used to simulate stress distribution, identify vulnerabilities, and predict the impact of environmental factors or physical damage on infrastructure. Such simulations underpin the decision-making processes for maintenance and operational optimization [66,100].

Simulation plays a pivotal role in the continuous updating of DTs to reflect changes in the physical system. Real-time data from sensors or imaging technologies can be fed into the virtual model, which is then recalibrated to ensure accuracy. For instance, stereo vision and other imaging methods can detect fatigue cracks in steel structures; this data can be integrated into FEM models to update the geometry and simulate the impact of the damage. These updates enable the DT to predict system behaviour dynamically, allowing stakeholders to address risks proactively. The seamless integration of real-world observations with simulation enhances the twin's ability to provide actionable insights and maintain relevance over time [66,101].

A core functionality of DTs is the predictive analysis, which is powered by simulation. For example, in earthquake engineering, FEM-based Incremental Dynamic Analysis (IDA) can be used to evaluate the response of structures to seismic forces, helping to assess collapse risks and improve resilience [102]. Similarly, in autonomous mining operations, FEM simulations can guide trajectory planning for excavators, optimizing load distribution and energy consumption while ensuring structural integrity. DTs enable testing and validation of strategies without disrupting real-world systems by simulating various operational

scenarios, thereby reducing risks and improving efficiency [101].

The incorporation of advanced technologies like AI and the IoTs further amplifies the role of simulation in DTs. AI-driven models enhance the predictive accuracy of simulations by learning from historical and real-time data, while IoT devices provide continuous streams of input to keep the DT synchronized with the physical system. For example, AI-based simulations in tunnel fire management can predict fire dynamics and inform emergency responses, demonstrating how DTs integrate simulations with real-world data to address complex scenarios. These hybrid approaches ensure that simulation remains central to the DT's functionality, providing a robust framework for real-time analysis and decision-making [103].

In conclusion, simulations are important to the creation and operation of DTs. They enable accurate representation, real-time updates, and predictive analysis, enhancing risk management and operational efficiency. Simulations ensure DTs remain vital tools for dynamic system optimization and informed decision-making by integrating real-time data and advanced technologies.

4.4. Discussion

The collective evidence from monitoring, modelling, and simulation research demonstrates that no single methodological strand can independently deliver the fidelity, responsiveness, and explanatory capacity required for contemporary DT deployments. Continuous monitoring through IoT or edge-enabled sensors supplies the temporal depth needed for early anomaly detection, yet without semantically enriched BIM-based models, these data streams lack the contextual structure necessary for lifecycle analytics. Conversely, high-resolution FEM simulations offer granular insight into structural behaviour and extreme-event response, but they require tight calibration to prevent model drift as conditions evolve. As a result, the most advanced DT implementations converge these techniques into hybrid workflows: live sensor data periodically updates probabilistic or AI-assisted models, which then drive high-speed surrogate simulations to support prescriptive, near-real-time interventions. Examples include adaptive tunnel lighting control, automated clearance checking for road widening, and fatigue prognosis in bridges.

Recent research continues to reinforce the methodological patterns outlined in this study, showing that DT development increasingly relies on multi-modal data acquisition to capture both as-built geometry and real-time structural or operational responses across bridges, tunnels, and buildings [86,87,90,104]. On the modelling and simulation side, BIM-centred virtual environments, in-situ behavioural models, high-fidelity FEM analyses, and ontology- or graph-driven representations are being used to encode geometry, physics, and semantic attributes for built assets at varying scales [87–89,105]. These developments confirm that current studies largely follow the same research trajectories previously identified [41,42,82,106].

However, these integrated pipelines expose persistent limitations. Data heterogeneity across sensing systems, latency introduced by cloud-centred architectures, and the computational burden of synchronizing

multi-scale simulations with sub-second refresh rates remain significant barriers. Addressing these challenges will require lightweight interoperability standards, physics-aware surrogate models hosted at the edge, and governance frameworks that incentivise structured data sharing across design, construction, and operation. Such advances are essential if DTs are to transition from isolated demonstrators to robust, infrastructure-scale decision-support systems.

A comparison across sectors further reveals clear alignments between methodological preferences and domain-specific priorities. Monitoring-intensive DTs are most mature in tunnels and bridges, where continuous sensing and real-time safety assurance are vital. Modelling-led approaches, especially those grounded in BIM and semantic ontologies, dominate buildings and construction management due to the need for rich data integration, facility operation, and energy optimisation. Simulation-driven DTs, frequently augmented with AI, optimisation algorithms, or multi-criteria decision analysis, are most advanced in roads and urban planning, where scenario testing, forecasting, and mobility analysis are central. Hybrid configurations are increasingly emerging at sectoral intersections, such as IoT-BIM integration for automated maintenance planning.

Overall, the evidence suggests that while each method addresses distinct operational requirements, meaningful progress in DT scalability and interoperability depends on their convergence. Hybrid workflows that combine real-time sensing, semantically coherent modelling, and predictive simulation offer the most promising pathway for achieving resilient, lifecycle-integrated DT ecosystems across diverse infrastructure contexts.

5. Sectors

DT applications vary considerably across both asset-based and process-based sectors, each presenting distinct use cases and integration challenges. For analytical clarity, this study classifies DT applications into six categories: tunnels, bridges, roads, buildings, construction management, and urban planning. The first four represent physical asset domains, whereas the latter two, construction management and urban planning, denote process- and system-oriented contexts in which DTs operate across multiple assets and lifecycle stages. This dual perspective enables the taxonomy to capture the full spectrum of DT deployment, from component-specific monitoring to project- and city-scale coordination.

DT applications vary considerably across sectors, each presenting distinct use cases, maturity levels, and integration challenges. Fig. 6 shows the distribution of research papers on DT applications across key infrastructure sectors, including buildings, construction management,

urban planning, roads, and multi-sector applications. Table 3 presents a comparative summary by sector to provide a clearer synthesis of the diverse applications and maturity levels of DT technologies across different domains. It contrasts typical use cases, enabling technologies, maturity, and prevailing implementation challenges. This overview highlights the variation in adoption readiness, for instance, buildings and tunnels exhibit more mature DT implementations due to well-defined sensing and modelling pipelines, while urban planning and road infrastructure still face hurdles related to interoperability and large-scale data fusion. Recognizing these differences is critical for tailoring research strategies, aligning stakeholder expectations, and guiding the development of sector-specific DT frameworks.

5.1. Tunnels

DT technologies have emerged as transformative tools in the monitoring, maintenance, and operation of tunnel infrastructure, offering a wide array of applications that enhance both efficiency and safety. Recent studies underscore the growing sophistication of DT implementations tailored to tunnel environments. For instance, Shen et al. [1] proposed an integrated DT framework aimed at optimizing tunnel lighting systems. Their approach balances energy efficiency with user safety by leveraging real-time data acquisition, advanced modelling, and analytics, significantly enhancing visibility while minimizing energy consumption. This comprehensive system exemplifies the dual potential of DTs to improve operational performance and environmental sustainability in tunnel management.

In a related advancement, Yu et al. [107] introduced a DT-based performance prediction model for highway tunnel pavements. Utilizing ML techniques, the model anticipates pavement deterioration over time, supporting lifecycle-based maintenance planning. This forward-looking approach enables infrastructure managers to optimize repair schedules and extend pavement lifespan, promoting cost-effective asset stewardship. Safety assurance remains a critical application of DTs in tunnels, especially in complex, long-span structures. Yu et al. [10] developed a DT-integrated structural health monitoring framework that combines real-time sensor data with simulation models to evaluate safety risks under variable load conditions. This system enhances risk assessment protocols and supports proactive safety interventions in challenging operational contexts.

In the domain of construction, DT technologies are also revolutionizing Tunnel Boring Machine (TBM) operations. For example, DT-enabled synchronization systems continuously adjust excavation parameters in response to real-time data on geological and structural conditions [10]. This responsive feedback loop reduces the likelihood of operational hazards and improves the precision and efficiency of tunnelling activities, highlighting the critical role of DTs in advancing construction automation and safety.

Collectively, these studies illustrate the multifaceted capabilities of DTs in tunnel engineering, from intelligent lighting and predictive maintenance to structural risk assessment and real-time construction control. These applications demonstrate how DTs are not only improving operational workflows but also contributing to safer, more sustainable tunnel infrastructure systems.

5.2. Bridges

Bridges have emerged as a key domain for the application of DT technologies, given their critical function within transportation networks and the high stakes associated with structural reliability. DTs offer new capabilities for real-time monitoring, predictive maintenance, and risk mitigation, making them invaluable tools for enhancing the safety, longevity, and efficiency of bridge infrastructure. A seminal contribution by Heng et al. [112] exemplified the power of DTs in structural health monitoring. Their framework enables the continuous assessment of bridge performance under varying environmental and load conditions

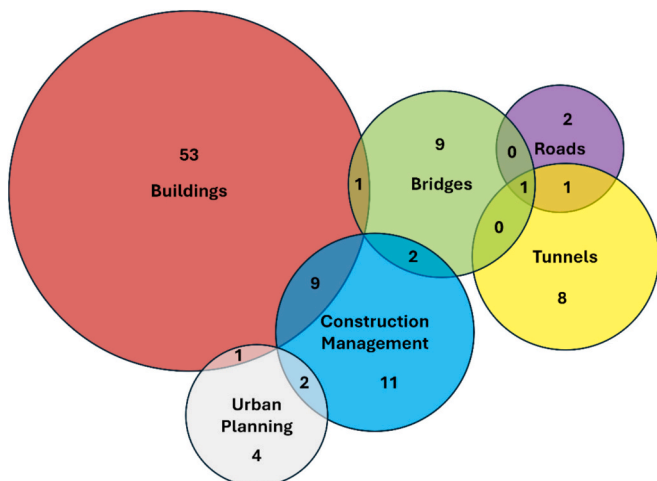


Fig. 6. Distribution of the selected studies by sector.

Table 3
Comparative overview of DT applications by sector.

Sector	Typical Applications	Dominant Technologies	Maturity Level	Key Challenges	Ref
Tunnels	Lighting control, performance prediction, safety monitoring	IoT, FEM, AI, BIM	Moderate to High	Sensor integration, real-time synchronization	[103,107,108]
Bridges	Structural health monitoring, seismic risk assessment	AI, FEM, ML, IoT	Moderate	High-fidelity modelling, predictive maintenance	[47,55,109]
Roads	Traffic analysis, clearance checking, maintenance	Edge AI, GIS, BIM	Emerging to Moderate	Data fusion, decision support integration	[55,97,107]
Construction Mgmt.	Resource allocation, scheduling, real-time visualization	BIM, IoT, AI	Moderate	Cross-phase continuity, real-time data capture	[53,65,75,93]
Urban Planning	Mobility simulation, energy optimization, land use planning	GIS, BIM, ML, Digital Simulation	Low to Moderate	Scalability, interoperability, standardization	[57,62,64,74,98]
Buildings	HVAC optimization, energy monitoring, heritage preservation	BIM, IoT, AI, VR/AR	High	Privacy, integration with legacy systems	[54,61,110,111]

by integrating FEM with real-time data streams from an extensive sensor network. This hybrid system supports predictive maintenance strategies and facilitates timely interventions, particularly under extreme weather scenarios, thereby safeguarding structural integrity over time. Recent work has proposed unified frameworks integrating monitoring, simulation, and analytics for bridge DTs, further advancing lifecycle management and interoperability [104].

Expanding the scope of DT utility, Jiang et al. [97] proposed a comprehensive DT framework tailored for highway bridges and overpasses, with a specific focus on clearance checking during road widening projects. Their methodology combines high-resolution DT models with advanced alignment fitting and online map data to conduct precise clearance analyses. This approach not only improves design accuracy but also eliminates the need for costly and labour-intensive field surveys, streamlining project delivery.

DT technologies have also been instrumental in evaluating the seismic resilience of complex bridge structures. Lin et al. [113] developed a collapse fragility assessment framework for cable-stayed bridges, underpinned by extensive FEM simulations. Their work provides detailed insights into structural vulnerabilities under seismic loads, supporting the development of targeted mitigation strategies to enhance infrastructure resilience in earthquake-prone regions. The integration of AI with DT platforms further amplifies their diagnostic and predictive capabilities. AI-driven models facilitate the early detection of anomalies within bridge structures, enabling rapid identification and remediation of defects before they escalate into critical failures [114]. This synergy between AI and DTs enhances asset reliability and supports more responsive infrastructure management.

These advancements illustrate the transformative impact of DTs on the lifecycle management of bridges. From real-time structural monitoring and automated clearance analysis to seismic risk assessment and intelligent fault detection, DTs are redefining how bridges are designed, operated, and maintained, ushering in a new era of data-informed, resilient infrastructure systems.

5.3. Roads

DT technologies are transforming road infrastructure by enabling more precise planning, responsive operations, and data-driven maintenance strategies. Their application spans the lifecycle of road networks, from initial design through to everyday management and long-term asset stewardship. A recent systematic review highlights similar advancements in DT applications for road pavement management, including predictive maintenance and lifecycle monitoring [105]. Jiang et al. [97] demonstrated how DTs can streamline road widening projects through virtual clearance checks. Engineers can pre-emptively identify and resolve spatial conflicts by constructing high-fidelity replicas of existing carriageways and adjacent structures. This method replaces laborious field surveys with rapid, in-silico analyses that uphold safety standards while accelerating design iterations.

Beyond geometric assessments, DTs have become essential in traffic

management and operational monitoring. A notable implementation combines a compact edge-AI module with a continuous DT feed to measure vehicle speeds and map flow dynamics across multi-lane free-ways [115]. This real-time system adapts signal timings or variable speed limits in response to emerging congestion, reducing incident risk, and improving throughput without human intervention. At the city scale, DTs are also guiding sustainable urban road planning. In one pioneering study, Jiang et al. [116] integrated microscopic traffic simulations, environmental modelling, and wind-flow analysis into a unified DT platform. Planners leverage this holistic view to balance mobility, air quality, and community impact, using multi-criteria decision algorithms to select alignments that minimize pollution and noise while meeting transport demand.

Finally, the maturation of DT-enabled predictive maintenance is transforming pavement management. Modern DT systems forecast wear patterns and optimize resurfacing schedules by fusing historical performance data, structural simulations, and live sensor inputs. This proactive approach extends pavement life, cuts maintenance costs, and enhances safety, delivering measurable benefits in both operational efficiency and network sustainability.

5.4. Construction management

DT technologies are reshaping construction management by enabling dynamic, data-driven oversight of complex project environments. Through their integration with tools such as BIM, AI, and the IoT, DTs are transforming how construction projects are planned, coordinated, and executed. Sun and Liu [117] proposed an advanced hybrid DT-BIM model that exemplifies this shift. Their framework enhances resource allocation by providing granular insights into material consumption and labour distribution across the construction site. The model promotes workflow efficiency and shortens project timelines by optimizing construction routing and automating core tasks such as scheduling and dispatching. The incorporation of AI ensures that the system remains adaptive to real-time changes, enabling responsive decision-making in dynamic construction environments.

Complementing this, Jiang et al. [97] presented a comprehensive DT framework that integrates real-time data streams with advanced visualization capabilities. Their approach supports proactive project tracking and issue resolution by unifying data from disparate sources and synchronizing it across construction activities. Stakeholders gain improved situational awareness, allowing them to identify delays, manage resource bottlenecks, and respond swiftly to on-site disruptions, ultimately enhancing overall project performance. DTs also play a critical role in improving on-site safety and compliance. DT frameworks can monitor environmental conditions, structural parameters, and equipment operations in real-time by capturing continuous data from embedded IoT sensors. Predictive analytics applied to these data streams enable the early identification of safety risks and the enforcement of regulatory standards. As highlighted by [103], this capability empowers construction managers to implement timely interventions, reducing the

likelihood of accidents and ensuring a safer working environment.

These developments illustrate the growing maturity and impact of DT applications in construction management. DTs not only enhance productivity and coordination but also elevate safety and quality assurance across all stages of project delivery by integrating real-time monitoring, predictive analytics, and intelligent automation.

5.5. Urban planning

Urban planning increasingly incorporates DT technologies to tackle the pressing challenges of sustainability and efficient resource management. For instance, Bortolini et al. [118] reviewed various innovative energy management tools that seamlessly integrate DT with multi-vector smart energy systems within urban environments. This sophisticated framework enhances the real-time coordination and management of various energy assets, thereby empowering local governments to meet their environmental goals while simultaneously achieving economic growth and energy efficiency.

Expanding on the potential of DTs in city management, Bigorra et al. [76] explored the conceptual foundations of an urban DT. Their study mapped existing DT initiatives across dimensions such as application areas, data inputs, processing mechanisms, and expected outputs. They also identified system requirements and proposed the core structural components of a city-level DT, outlining the associated benefits, existing gaps, and principal challenges that need to be addressed. Similarly, Liu et al. [41] reviewed DT modelling strategies for the operation and maintenance of urban infrastructure, emphasizing that large-scale implementation continues to be hindered by persistent issues in data integration and governance.

Furthermore, DT technologies have found relevance in the preservation and management of historical sites. These technologies help safeguard cultural heritage by blending advanced digital modelling techniques with real-time monitoring systems. This innovative application ensures that historical landmarks are not only protected but are also integrated into the framework of sustainable urban development, allowing cities to celebrate their rich cultural narratives while promoting ecological responsibility [76].

5.6. Buildings

The integration of DT technology in the management of buildings has primarily aimed to enhance operational efficiency and promote sustainability. Recent reviews further consolidate the role of DTs across the built environment, emphasizing common challenges related to interoperability, standardization, and lifecycle data management [35]. In their research, Alavi et al. [119] introduced a comprehensive DT framework specifically designed for smart buildings, which combines BIM with IoT technologies. This innovative approach enables real-time monitoring of building systems, enabling predictive maintenance to address issues before they escalate. The framework not only facilitates the continuous tracking of energy consumption and resource usage but also incorporates advanced algorithms for energy optimization, ensuring that buildings operate at peak efficiency. Moreover, it prioritizes occupant comfort by monitoring environmental factors such as temperature, air quality, and lighting, adjusting them dynamically to meet the preferences of individuals within the space. Their model represents a significant step forward in the evolution of building management practices by addressing these critical challenges, ultimately contributing to more sustainable and user-friendly environments [119,120].

DT technologies are increasingly being integrated into the preservation of historical buildings. Fan et al. [121] developed an innovative DT-based framework specifically designed for the protection and enhancement of historical sites. This comprehensive system employs advanced 3D modelling techniques to accurately represent the physical intricacies of these structures. In addition, it utilizes knowledge graphs

to establish connections among various data points related to the site's history, architectural features, and current condition. Real-time data integration plays a crucial role in this system, as it allows for continuous monitoring of environmental risks such as structural damage, moisture levels, and other potential threats to the integrity of the buildings. The framework can propose tailored preservation strategies by analysing this live data that ensure the longevity of these cultural treasures.

Moreover, the integration of AI and ML into DT systems has significantly elevated their functionality in the management of building environments. These technologies facilitate the dynamic control of energy systems, leading to optimized energy consumption and improved sustainability within historical structures. These advanced DT systems promote not only the preservation of the building's heritage but also enhance the comfort and experience of its visitors by adapting responses to real-time environmental changes, such as fluctuations in temperature, humidity, and occupancy levels. The fusion of DT technologies with AI represents a remarkable stride toward safeguarding our historical architecture while ensuring its relevance in the modern world [122].

The integration of DT with advanced technologies such as AR/VR has further enriched building management applications. For instance, DT-enabled visualization tools have facilitated interactive simulations for facility management and occupant training, providing valuable insights into building performance and maintenance needs [123]. These developments underscore the transformative potential of DT in creating smart, sustainable, and resilient building environments.

5.7. Discussion

Across the six domain-oriented case studies, tunnels, bridges, roads, construction management, urban planning, and buildings, the evidence demonstrates that DTs deliver a largely consistent value proposition: real-time condition awareness, predictive maintenance, and data-driven optimisation. Yet the sectoral analyses also reveal nuanced distinctions that must inform future research and deployment strategies. Safety-critical assets with low inspection accessibility (e.g., tunnels and long-span bridges) emphasize high-fidelity sensing and physics-based simulation to mitigate catastrophic risks, whereas energy-intensive facilities, such as smart buildings, prioritise IoT-enabled monitoring and AI optimisation to reduce operational carbon emissions. Transport-network DTs (roads and city-scale mobility platforms) showcase the integrative power of edge AI and multi-criteria decision frameworks for coordinating traffic, environmental quality, and land use, but they also expose acute interoperability gaps when fusing heterogeneous data streams. Construction-phase DTs highlight workflow orchestration benefits, yet their effectiveness still hinges on seamless phase transitions into operation and maintenance, a maturity rarely achieved in practice. Collectively, these findings underline that while core DT technologies are transferable, implementation pathways must be tailored to sector-specific risk profiles, data granularities, and lifecycle priorities. Bridging these silos will require harmonised ontologies, lightweight middleware for cross-platform data exchange, and demonstrators that quantify return-on-investment over entire asset lifecycles. Only then can DTs progress from isolated exemplars to an integrated, systemic infrastructure intelligence layer.

Recent publications further reinforced the trends identified in this study, demonstrating that developments in DT research continued to evolve along the same trajectories previously reported in the literature. For buildings, recent work [124] highlighted the growing shift toward occupant-centric DTs, emphasizing user interaction, comfort monitoring, and data-driven operational decision-making, fully aligned with earlier work on DT-enabled building management. Similarly, another study [83] proposed a BIM-IoT-GIS integrated DT framework for relocatable modular buildings, mirroring established trends in lifecycle information integration, real-time monitoring, and coordinated facility management. In the infrastructure domain, research in highway DTs developed an advanced spatial-temporal graph network for pavement

prediction within highway DTs, reaffirming the increasing use of AI-driven analytics for preventive maintenance and long-term asset management [125]. Complementary work presented a unified framework for bridge DTs that synthesised sensing, modelling, and cyber-physical interfaces, demonstrating continued emphasis on structural condition assessment, automated inspection, and lifecycle structural health monitoring [104]. Additionally, further studies reviewed DT applications across urban-scale infrastructure systems, again underscoring multi-scale modelling, data fusion, and operation and maintenance-oriented decision support [41]. These recent studies confirmed that research in buildings, roads, and broader infrastructure consistently followed the same conceptual and technological directions highlighted in our review, namely, integrated data environments, AI-supported analytics, real-time or near-real-time synchronization, and lifecycle-focused operational intelligence, demonstrating that the trends identified in our paper remained valid and up to date [43,90,104–106].

Across sectors, a clear methodological pattern emerges. Monitoring-centric DTs dominate safety-critical assets such as tunnels and bridges, where real-time sensing and FEM-based modelling ensure early anomaly detection. In contrast, modelling-intensive DTs, particularly BIM- and ontology-based frameworks, are most effective in buildings and construction management, where data richness and interoperability enable energy optimisation and workflow coordination. Simulation-driven DTs gain prominence in road and urban planning applications, supporting predictive traffic control and scenario-based policy testing. Despite these differences, the most resilient implementations integrate at least two methods, balancing real-time responsiveness with semantic coherence and predictive depth. Common challenges across sectors include inconsistent data schemas, fragmented governance, and weak continuity between construction and operation. Drawing from recent reviews in each domain, it becomes evident that sector-specific lessons, such as the value of ontology-based modelling in buildings or edge-AI monitoring in transport, can inform unified frameworks for multi-asset digital infrastructure management.

6. Phases

The flexible nature of most DTs means that they can be deployed during any phase in the infrastructure lifecycle of a system. These can be further narrowed down to representation of the past (the design of the project), the present (the project's construction) and its future (any necessary operation and maintenance post creation). Fig. 7 is a Venn Diagram breaking down how the literature is separated between each lifecycle phase and their respective intersections. Each stage of these processes has different requirements, use cases and specifications that must be considered before deployment; however, the surveyed literature has shown that incorporating DTs are beneficial to the end user, saving

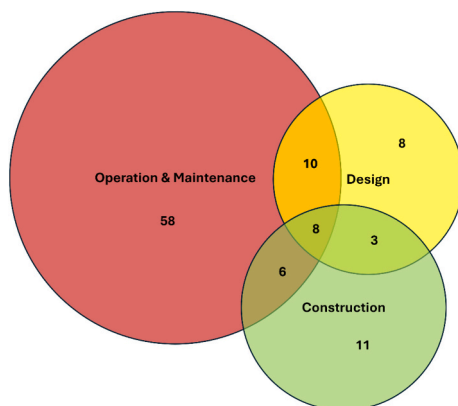


Fig. 7. Distribution of selected studies across the three primary phases of the infrastructure lifecycle.

time, resources, and money as a result. While DTs are easy to use and conceptualise, their design is strengthened depending on how they are tailored to the problem they are trying to solve. These papers show how DTs excel in specific areas when applied and how they approach their integration into the project. Table 4 summarizes the characteristics of DT use in the **design**, **construction**, and **operation and maintenance** phases to clarify the distribution and focus of DT applications across the infrastructure lifecycle. While the majority of studies focus on operation and maintenance, where the return on investment is more immediate, DT applications in earlier phases are gaining traction, particularly in sustainability assessment and construction site optimization. However, phase-spanning continuity remains a challenge, with significant information loss occurring at the design–construction–operation transition. Future efforts should aim to develop integrated DT pipelines that persist and evolve across the entire lifecycle.

6.1. Design

DTs can be integrated into infrastructure design even before construction commences. The 29 papers in this section highlight a series of common design themes in which DTs have been effectively deployed. For example, Alnaser et al. [59] explored a structural equation modelling framework to evaluate how well DTs are integrated into BIM. The study provided a quantitative metric for weighting these factors and analysing their correlations by categorizing the results into Key Success Factors (KSFs) and sustainability parameters. This led to a focus on cost optimization and resource management as key priorities, along with other KSFs that businesses should adopt.

A primary reason for employing DTs early in the development process is their ability to plan the entire lifecycle of a building and establish boundaries and parameters to guide the project. Kang et al. [96] proposed a framework that monitors the environmental performance of buildings and defines the requirements necessary to enable real-time connectivity and predictability. This relationship is demonstrated through the examination of use cases and the development of a model that addresses existing research gaps. Ward et al. [75] also explored the challenges associated with designing real-time connectivity, focusing more specifically on live-streamed data and the provision of continuous access and storage for data processing. As data demands increase exponentially, these studies, along with [127], highlight the need for systems capable of modelling real-world data from the outset of the lifecycle. Doing so helps prevent issues from compounding over time, ultimately saving both time and resources.

These works also highlight environmental concerns as a key factor in incorporating DTs. Ding et al. [127] and Zhang et al. [103] explored recent advances linking DTs to fire safety, leveraging sensor data and AI to develop systems that can identify fire locations and ensure building designs comply with up-to-date fire safety standards. This includes both the prevention of future fire hazards and the minimisation of their impact if a fire does occur. Their approach focuses on using YOLO to track evacuees and assess how effectively they can reach the nearest fire exits, as well as to analyse the location and intensity of fire outbreaks in real-time.

In addition, Naser [52] applied AI and ML to develop a virtual assistant capable of detecting vulnerabilities in concrete mixtures, including fire-induced spalling, through a series of case studies. Li et al. [94], by contrast, used a chiller as a physical object to demonstrate their proposed integration of static information, physical mechanisms, and interaction patterns to enhance safety, comfort, and energy savings. Desogus et al. [60] addressed these key performance indicators (KPIs) by examining thermal comfort conditions and how they can be optimized through BIM integration.

Kamari et al. [128] and Koo et al. [78] also employed neural networks and ML to predict the impact of natural disasters and operational uncertainties on construction sites, assessing the threats these pose to future building projects. These approaches offer the advantage of being

Table 4
Comparative summary of DT applications across lifecycle phases.

Phase	Typical Objectives	Main Methods/ Technologies	Benefits	Challenges	Ref
Design	Lifecycle planning, environmental analysis, risk prediction	BIM, AI/ML, simulation, MCDM, GIS	Improved planning; early risk mitigation; sustainability	Low adoption rate; limited standardization across design platforms	[94,96,126]
Construction	Real-time progress monitoring, resource management	BIM, IoT, sensor data, data mining, image processing	Improved site coordination; safety; reduced delays	Weak data continuity to operation and maintenance; costly retrofitting of DTs	[49,53,79,101]
Operation and Maintenance	Fault detection, predictive maintenance, energy optimization	AI/ML, IoT, FEM, BEM, GNN, UAVs, AR/VR	Energy efficiency; structural health monitoring; resilience	Retroactive DT creation; interoperability; model drift over time	[12,98,107]

non-intrusive methods for contour mapping, projection, model development, and benchmarking, enabling future optimisation in later lifecycle phases. Jiang et al. [116] integrated DTs with Multi-Criteria Decision Making (MCDM) and GIS to develop an eco-friendly road planning scheme based on existing land use, traffic congestion, air quality, and noise levels, aimed at minimizing environmental impacts on a broader scale beyond just heat management.

Zhou et al. [62] demonstrated how the simulation of social attributes in urban design and governance can be harnessed and integrated into DT research and development. In contrast, Daniotti et al. [14] adopted the inverse approach of Alnaser et al. [59] by using BIM to inform the design of a DT for construction. This was achieved through a framework of ontologies that support activities commonly associated with the modern renovation of residential buildings.

6.2. Construction

With only 28 papers, this is the least explored of the three major areas in the infrastructure lifecycle. These studies typically focus on topics such as reviews of current construction techniques (e.g., the general contractor's perspective or drivers for DT adoption) or model transformation methods (e.g., semi-automatic geometric DTs and panorama-to-model registration), highlighting how to translate theoretical concepts into practical, real-world applications.

One of the most prominent strands in construction-related DT research involves the use of BIM to design frameworks [99], aiming to enable real-time data transmission in contexts where no prior examples existed to support stakeholder engagement. Emphasizing real-time decision-making allows the DT to auto-update in response to newly entered construction data, enabling it to react to adverse conditions and support the immediate application of newly developed frameworks. These frameworks either build on existing models by integrating complementary techniques such as IoT or data mining (DM), or they propose entirely new conceptual approaches, pushing the field forward and establishing new standards for future research.

Both review papers in this section examine the state of research in the Architecture, Engineering, Construction, and Facility Management (AEC-FM) industry [58,129]. They cover information management research ranging from model-based approaches to building systems and their interaction with smart cities, as well as the automation solutions that emerge from these interactions. Both papers focus on keyword trends in studies published between 2013 and 2022, identifying deficiencies in areas such as "Integration Standards," "Predictive Maintenance," and "Facility Maintenance." They advocate for a shift from theoretical research to practical implementation, aiming to transform DTs into built entities that support more effective decision-making.

Building on this overarching theme, references [46, 65] offered overviews, critiques, and analyses of prior literature in the construction industry. While Esmaeili and Simeone [46] did not specify the number of papers included in their review, Cheng et al. [65] and Opoku et al. [130] examined non-exclusive papers on digital applications up to 2023. Their reviews highlighted the gap between the development and utilization of certain technologies and explored how concept-oriented and

production-driven drivers had been adopted and identified across the field.

Liu et al. [111] and Gao et al. [66] demonstrated the utility of applying DTs to steel structures, while Jiang et al. [101] focused specifically on cranes, illustrating how DTs can improve the detection of potential buckling in steel caused by unsafe hoisting practices. The examples presented in these studies range from monitoring structural behaviour and remeshing welded joints after fatigue cracks emerge, to conducting continuous safety assessments based on real-time data and observations. One outcome is a Support Vector Machine (SVM) algorithm capable of predicting overall structural safety levels, alongside another algorithm designed for crack detection, identification, quantification, and registration, achieving 91.9% accuracy in identifying specified welds in bridges. DTs are shown to enable safer structural design, better synchronization between digital and physical assets, and the development of novel safety evaluation methods.

Lin et al. [100] and Li et al. [109] also focused on steel bridges, addressing long-term challenges such as deformation and the risk of collapse. Both studies examine the effects of deformation over time using FEM and incremental dynamic analysis, with an emphasis on detection, estimation, and corrective strategies. Their findings are compared with theoretical and practical outcomes, validating the feasibility of these approaches for real-world bridge assessments.

More general management approaches are discussed in [53,76], each capable of supporting decision-making across an asset's life cycle. These papers explore how BIM-based processes can be integrated within DT frameworks, supported by both theoretical models and practical data analysis. The results indicate improvements over conventional implementations in areas such as computational performance, reduction of initial technology deployment costs, identification of resource constraints, and effective allocation of existing requirements.

6.3. Operation and maintenance

Once the building has been constructed, DTs can be applied throughout the remainder of its lifecycle, encompassing both present and future operations. This stage comprises 82 papers, making it the largest of the three sections. Siccardi et al. [131] explored modern trends in facility management (FM), discussing technologies such as BIM, IoT, AI, and ML, while focusing on how DTs integrate with these technologies to enhance FM applications. Cheng et al. [55] addressed the role of DTs in the emergency management of civil infrastructure, proposing a framework that applies DTs across the mitigation, preparedness, response, and recovery phases, highlighting their potential to address existing limitations and influence recovery trends. Spudys et al. [61] identified a gap in the declaration of energy performance certificates and the integration of BIM and DT technologies. They proposed a framework in which operational indicators feed into a DT environment supported by smart sensors, real-time measurements, and a BIM-based platform.

Using graph representation, Gnecco et al. [68] introduced a novel approach to DT development by integrating monitoring data and occupant surveys to map the relationships between building agents for

experimental purposes. Zhao et al. [73] also explored DTs in conjunction with BIM within the context of near net-zero energy buildings. Their work employs laser scanning to develop Building Energy Models (BEMs) of existing structures, allowing for the identification and evaluation of retrofitting schemes. This demonstrates how DT-designed buildings of the future can achieve greater energy efficiency than their current counterparts.

Hasamo et al. [91] is one of four reviewed studies that discuss the application of neural networks for the operation and maintenance of DTs. Their proposed framework leverages the emerging use of DTs in the facility management sector, particularly through BIM and IoT, to improve maintenance strategies for building systems, including the development of HVAC systems, which is also a central theme in [56,70]. The field of ML for predictive maintenance is reviewed, with artificial neural networks, SVMs, and decision trees selected to forecast future conditions.

In light of the impacts of natural disasters and the demand for rapid emergency responses, Cheng et al. [55] emphasized the importance of computer vision and deep learning in DT applications. Their work highlights the broad use of UAV-based remote sensing for data collection across civil infrastructure, contrasting with studies like Zheng et al. [78], which focus on post-incident monitoring. In contrast to other DT construction studies related to energy, such as [15,61,68,71,73,74,132,133], this approach deploys neural networks specifically to enhance environmental comfort and achieve energy savings.

Graph Neural Networks (GNNs) establish patterns among various systems in the built environment, such as wearable sensors, different users, and support equipment. A key advantage of this method is its adaptability, as the models can evolve and improve over time in response to ongoing modifications. Zhao et al. [77] integrated Artificial Neural Networks (ANNs) with a DT operation and maintenance model to establish a comprehensive DT framework. Neural networks and decision trees were used to develop a model encompassing geometric, physical, and behavioural aspects. This model provides services such as predicting building conditions, generating residual lists, and analysing the causes and effects of structural deterioration and risk. These findings align with studies such as [13,68,71,73,132], which illustrate the intersection of operation and maintenance in buildings and emphasize the potential of DTs in enhancing building maintenance strategies.

Bridge engineering remains a prominent use case for DTs beyond the initial design and construction phases. Several studies [11,12,92,114] have examined current DT implementation methods and how recent innovations have advanced state-of-the-art practices across various aspects of bridge development and maintenance. These include literature reviews, structural health monitoring techniques analysing stress distribution and fatigue, and frameworks aimed at enhancing the efficiency and safety of bridge operations and maintenance. Similarly, Yu et al. [10,107] and Machado et al. [108] pursued comparable goals in the context of tunnels, identifying structural defects, predicting performance over time, designing proactive maintenance systems, and proposing a general DT-based analytical framework for tunnel operations and upkeep.

6.4. Discussion

Across the lifecycle-oriented literature, DTs clearly demonstrate value at every phase, yet their deployment remains uneven and largely siloed. Most studies focus on operation and maintenance, where real-time sensing and predictive analytics deliver immediate returns in energy optimisation, fault detection and asset health management. By contrast, only 29 papers address the design stage and 28 papers the construction stage, signalling that early-phase integration of DTs is still emerging. This imbalance has practical consequences: without robust design-time modelling and construction-time data capture, many twins must be “retro-fitted” once the asset is in service, limiting their fidelity

and increasing long-term costs.

These trends continued in papers outside the original range of data collection [40–42]. Design-focused papers such as [134] provided an up-to-date review of construction quality management through a layered framework that identified inference approaches and linked them to model functions across specific construction phases. Similarly, [135] examined how common architectures were designed and how DTs fit into the lifecycles of construction projects, highlighting implementation challenges such as improving data quality and security while developing robust integration methodologies. The construction sector also recently produced literature such as [136], which outlined methodologies for managing built environments through DTs, BIM, IoT, and AI, while [59] pinpointed key success factors and parameters in successful BIM–DT deployment and their correlations. Health monitoring remained an important focus in the operation and maintenance sector, with [137] showcasing a series of practical applications and use cases for DTs in bridges and structures, as well as the challenges posed by existing bridge DTs in the industry. In the same domain, [84] developed a framework that also accounted for IoT, AI, and BIM, while outlining the limitations of the current field and recommending mobile applications that could make proposed systems more aligned with the expectations of DTs in enterprises and national economies.

The evidence suggests that the continuity of information, carrying geometry, semantic data, and sensor context from BIM-based design through construction loggers to live operation and maintenance dashboards, is critical but currently represents the weakest link. Studies that do attempt such continuity report smoother hand-offs and fewer data-reconciliation errors, yet they also highlight persistent hurdles around interoperability, version control and governance of multi-disciplinary data flows. Moving forward, research should therefore prioritise phase-spanning frameworks that (1) embed DT thinking in conceptual design, (2) mandate structured, standard-compliant data collection during construction, and (3) enable automatic model evolution once the asset becomes operational. Achieving this end-to-end integration will not only unlock the full predictive power of DTs but also create a virtuous feedback loop in which insights from operation and maintenance continuously refine design heuristics for the next project cycle.

7. Scales

Scale refers to the scope or extent of the DTs coverage, ranging from small, detailed components (object scale) to entire infrastructure assets (asset scale) and up to large urban environments (city scale). Each scale presents unique requirements and challenges, influencing the DT's structure, data needs, and functionalities. At the object scale, DTs focus on specific components or subsystems within larger assets. The asset scale addresses the entirety of an infrastructure asset, such as a full bridge, building, or tunnel. This scale involves integrating data from multiple components to provide a comprehensive view of the asset's performance, structural health, and operational status. At the city scale, DTs extend to urban environments, covering extensive networks of interconnected infrastructure such as road systems, utilities, or entire cities. City-scale DTs are typically employed for high-level planning, resource management, and emergency response, and they require robust data aggregation and integration from various sources.

This section identifies 20 papers focused on object-scale DTs, 66 papers on asset-scale DTs, and 7 papers on city-scale DTs, as shown in Fig. 8. Additionally, 3 papers explore the overlap between asset and city scales, and 7 papers explore the overlap between asset and object scales. The predominance of research at the asset scale reflects the current emphasis on improving individual infrastructure assets, while city-scale applications remain an emerging area with growing interest. The following sections synthesize key findings from each category, highlighting advancements, challenges, and trends in DT research at different scales. Table 5 compares object-, asset-, and city-level DTs to illustrate how the scale of implementation influences the objectives,

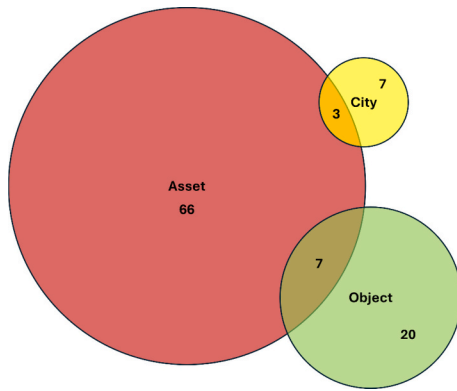


Fig. 8. Distribution of reviewed papers based on three scales of DT implementation.

methods, and complexity of DT applications. Object-scale DTs are typically focused on detailed diagnostics and component-level analytics, while asset-scale twins integrate broader operational data to support lifecycle performance. City-scale DTs are emerging and present immense potential for coordinated urban management but also require robust data infrastructures and governance frameworks. Understanding these scale-specific characteristics is essential for designing interoperable and scalable DT ecosystems.

7.1. Object scale

This section focuses on the object scale, a domain where DTs are applied to monitor, optimize, and analyse individual structural components, systems, or processes. The object scale provides granular insights into specific elements, such as building components, equipment, and structures, making it a critical level for detailed modelling and real-time interventions. The reviewed papers demonstrate the versatility of DTs in structural safety, predictive maintenance, construction processes, and spatial modelling. DTs have shown significant potential in enhancing structural safety and monitoring capabilities. Liu et al. [111] introduced a multidimensional DT model for prestressed steel structures, utilizing real-time monitoring and predictive analytics to improve safety assessments. Similarly, Gao et al. [66] proposed a vision-based framework to track fatigue-induced geometric changes in steel joints, demonstrating enhanced monitoring precision.

DTs also play a crucial role in predictive maintenance methodologies. Hosamo et al. [91] developed a framework combining BIM, IoT, and ML for air handling unit maintenance, achieving significant energy savings and operational efficiency improvements. Gourabpasi and Nik-Bakht [70] proposed a BIM-based approach to enhance fault detection in HVAC systems, addressing limitations in sensor and contextual data availability. Zhang et al. [139] integrated IoT and ML models to optimize the operation of TBM, improving performance metrics such as penetration rates and energy consumption.

In infrastructure management, Hosamo and Hosamo [114] reviewed the integration of 3D laser scanning and DTs in bridge maintenance, identifying challenges like standardization while proposing conceptual frameworks for improvement. Jiang et al. [97] demonstrated the application of DTs for underpass clearance checking, providing a cost-effective alternative to field surveys during highway widening projects. DTs have also been employed to investigate structural failures. Real-time monitoring of indoor environments benefits significantly from DT technologies. Hu and Assaad [110] proposed a BIM-enabled framework combining IoT, robotics, and LiDAR for monitoring indoor air quality and visualizing spatial data in real-time. This integrated approach enhances facility management by identifying pollutant sources and assessing air quality with high spatial accuracy.

In the construction domain, Li et al. [67] introduced a DT-driven virtual sensor framework for monitoring and diagnosing faults in dredging equipment, improving safety and operational reliability. Du et al. [138] applied DT technology to cold-formed steel members, integrating 3D laser-scanned geometries into FEM for improved structural analysis and deformation prediction. DTs also enable innovative approaches to architectural design. Kalantari et al. [64] developed Ph2D, a hybrid tool for synchronizing physical prototypes with digital analysis, enhancing design experimentation and real-time simulation capabilities for architects and designers.

Lastly, advancements in spatial and image registration have been made using DTs. Wei and Akinci [79] introduced a hierarchical registration method combining semantic segmentation and reprojection error minimization, achieving sub-meter accuracy for localizing panoramic images in real-world settings. At the object scale, DTs provide capabilities in monitoring, predictive maintenance, safety analysis, and spatial modelling. They offer precise, data-driven insights into the behaviour and performance of individual components and systems. These applications demonstrate the potential of DTs to improve workflows and decision-making processes across various domains, setting the foundation for broader asset- and city-scale implementations.

7.2. Asset scale

This section explores the role of DTs at the asset scale. At this level, DTs represent and manage individual buildings, bridges, tunnels, and other infrastructure systems, focusing on their lifecycle, operational efficiency, and maintenance. The reviewed literature highlights the evolution of DT applications in managing assets, integrating advanced technologies, and addressing specific challenges such as energy efficiency, structural health monitoring, and predictive maintenance.

DTs are increasingly adopted to manage the entire lifecycle of buildings, from design and construction to operation and maintenance. Early implementations, such as Peng et al. [13], demonstrated how DTs can integrate static and dynamic data to support real-time facility management in hospital buildings, improving fault detection and maintenance. Zhao et al. [73] employed 3D laser scanning and BIM to create digital energy models, enabling evaluations of retrofitting schemes for nearly zero-energy buildings, which showed significant

Table 5
Comparative summary of DT applications by scale.

Scale	Scope	Typical Use Cases	Enabling Technologies	Benefits	Key Challenges	Ref
Object	Individual components or subsystems	Sensor diagnostics, HVAC units, steel joints, fatigue/crack detection	IoT, ML, FEM, LiDAR, AR/VR	High precision monitoring; localized fault detection	Limited generalizability; data fragmentation	[66,67,138,139]
Asset	Entire infrastructure entities (e.g., bridge, tunnel, building)	Structural health monitoring, facility management, predictive maintenance	BIM, AI/ML, IoT, GIS, BEM	Lifecycle integration; operational efficiency	Integration complexity; data interoperability	[11,51,126]
City	Urban infrastructure systems or networks	Mobility planning, energy optimization, land use, emergency response	GIS, edge AI, multi-agent simulation, IoT	Holistic decision-making; supports smart city development	Scalability; governance; privacy and standardization issues	[74,98,116,133]

reductions in energy costs and carbon emissions. Hakimi et al. [63] focused on infrastructure predictive maintenance by integrating data fusion techniques with openBIM, GIS, and extended IFCs.

The importance of operational monitoring has also been emphasized. Spudys et al. [61] proposed a framework for real-time operational energy performance assessment, while Desogus et al. [60] integrated sensor data with BIM to optimize energy use and thermal comfort. Similarly, Ni et al. [132] highlighted the use of advanced ML models, such as Temporal Fusion Transformers, for accurate energy consumption predictions. Yoon and Koo [4] advanced this by proposing in situ model fusion techniques to continuously verify and calibrate DTs for long-term building operations. Sivori et al. [95] leveraged DTs integrated with vibration data for seismic monitoring and health assessment of historic structures, exemplified by the Consoli Palace case study in Italy. These advancements are complemented by novel visualization methods. For example, Shahinmohadam et al. [49] integrated BIM, IoT, and real-time thermal data into a VR platform for visualizing indoor comfort levels.

DTs are also transforming construction and infrastructure management by enabling real-time monitoring and decision-making. Pal et al. [93] introduced a framework for real-time construction control using vision-based progress monitoring, while Sun and Liu [53] demonstrated the integration of DT and BIM to enhance resource management in construction projects. Zhao et al. [140] proposed a DT framework for optimizing the hoisting of prefabricated components, demonstrating improvements in real-time monitoring and planning. Esmaeili and Simeone [46] extended these applications to construction DTs, showcasing their ability to optimize resource allocation and collaboration. In addition, Siccardi and Villa [131] emphasized the rapid adoption of DT technologies like IoT and blockchain for facility management.

In bridge and tunnel management, DTs play a vital role in structural monitoring and predictive maintenance. Broo et al. [126] presented a bridge DT that integrates physical, cyber, and service layers, while Honghong et al. [47] proposed a DT-enhanced BIM framework for lifecycle management in bridge engineering. Yang et al. [12] expanded on these ideas by identifying gaps and proposing a lifecycle-oriented reference framework for bridge DTs.

In tunnel management, Li et al. [9] developed a high-precision tunnel modelling method combining image recognition and point cloud processing. Machado and Futai [108] introduced an inspection DT framework to predict intervention needs and optimize maintenance. Gao et al. [11] addressed communication challenges in bridge maintenance using an AIoT-informed DT framework, enhancing operational efficiency and fault tolerance. DTs have also been applied in protecting heritage assets. Liu and Wang [141] proposed a comprehensive DT-based framework for monitoring historical sites, integrating real-time data to inform protection strategies.

Advanced integration frameworks are expanding the application of DT applications. Wang et al. [15] extended BIM's capabilities by integrating SensorML for real-time environmental monitoring. Li et al. [94] proposed a unified DT framework combining static data, physical mechanisms, and interaction patterns, validated through a practical application in a chiller system. Finally, Opoku et al. [51] demonstrated a DT for real-time monitoring in a university library, integrating IoT and BIM to enhance facility management and occupant comfort.

7.3. City scale

This section focuses on the broader deployment of DT technologies in city-scale systems. City-scale DTs provide a comprehensive approach to urban management, integrating diverse data sources and decision-making frameworks to address complex challenges in urban planning, resource management, and governance. This section examines key research in the application of DTs at the city scale, with particular attention to energy systems, urban planning, water management, and the challenges of implementation.

DTs have emerged as a pivotal tool for managing multi-vector energy systems in smart cities. O'Dwyer et al. [74] introduced an energy management tool that integrates DT technology to optimize energy coordination, forecasting, and control. The system demonstrates its applicability through case studies in Greenwich, London, effectively managing communal heating and electric vehicle charging systems. This research highlights the potential for city-scale DTs to align energy operations with environmental, economic, and resilience objectives.

City-scale DTs are increasingly being applied to urban decision-making and planning. Zhou et al. [62] explored the integration of evolutionary game theory and reinforcement learning into DT city frameworks to support governance and community regeneration. Their approach simulates social interactions and governance dynamics, offering insights validated through a case study in Shanghai. Similarly, Jiang et al. [116] proposed a DT framework combining MCDM and GIS to optimize sustainable urban road planning. Their approach emphasizes the potential for eco-friendly and people-centric urban infrastructure development by addressing factors such as traffic congestion, air quality, and land use.

Efficient management of urban water systems has also been a focus of city-scale DT research. Li et al. [133] applied DT technology to urban drainage systems, employing a five-dimensional model and the Multi-Level Dynamic Priority and Importance Scheduling (MDPIS) algorithm to enhance emergency drainage and water-energy management. Wu et al. [98] developed a high-fidelity DT framework that integrates physics-based and data-driven models for anomaly detection and localization for Smart Water Grid management. Tested in Singapore, this framework demonstrates the capability of DTs to support proactive and efficient urban water management.

Despite these advancements, several challenges remain in the design and implementation of city-scale DTs. Lei et al. [57] conducted a systematic review and Delphi expert survey, identifying 14 technical and 9 non-technical challenges, including interoperability issues and the lack of robust business models. Their findings underscore the importance of overcoming these barriers to fully realize the potential of city-scale DTs.

City-scale DTs offer potential for urban management by integrating advanced technologies, data-driven approaches, and interdisciplinary methods. The research discussed in this section demonstrates their effectiveness in optimizing energy systems, improving urban planning, and managing water resources. However, addressing the identified challenges will be critical for scaling up these applications and achieving sustainable urban development.

7.4. Discussion

The literature on scale reveals a clear maturity gradient: most studies concentrate on the asset level, where integrating BIM models with real-time sensor feeds yields the most immediate return in terms of structural-health insights and facility performance optimisation. Object-scale work excels at high-resolution diagnostics, e.g., crack propagation in steel joints or chiller-level energy control but remains fragmented and largely confined to single-equipment case studies, limiting the generalisability of findings across whole assets or portfolios. At the opposite extreme, only 7 city-scale twins tackle system-of-systems challenges such as multi-vector energy balancing, urban drainage, or mobility planning; their slow uptake is attributed to interoperability gaps, heterogeneous data governance regimes and the absence of sustainable business models.

Recent studies published beyond the initial review period further reinforced these trends in scale, where asset scale remained the primary focus in DT research [43,105,106]. For example, recent work on LLM-enabled human-robot interaction in facility inspection demonstrated how DT-centred workflows could be used to orchestrate autonomous data collection and close the feedback loop between physical building systems and their DTs [142]. Similarly, the introduction of natural language interfaces such as ChatTwin illustrated a growing effort to

improve the accessibility and usability of asset-level DTs by enabling operators to retrieve DT data, trigger DT-supported analyses, and execute operation and maintenance tasks through natural language-based interaction, thereby lowering adoption barriers for users with limited digital skills [143,144]. At a broader scale, recent studies on highway infrastructure, such as network-wide pavement condition prediction using spatial-temporal graph models [125], and regional data-quality assessment using graph-based learning [145], extended DT analytics beyond single assets toward network-scale management. Likewise, Mok et al. [146] investigated the automation of defect detection and tracking pipelines for road assets. These studies did not yet represent fully integrated city-scale twins, but they demonstrated the progressive scaling of DT analytics to multi-asset, network-wide systems with heterogeneous, spatially linked data. Taken together, these papers reinforced the overall trajectory observed in the review: the field was most mature at the asset level, object-scale DTs remained specialized and fragmented, and system- or city-scale deployments emerged more slowly due to the computational and organizational challenges of integrating heterogeneous asset-level twins.

Taken together, these trends suggest that DT research advances “inward-out”: investigators first perfect methodologies on discrete components, then extend them to single assets, while city-scale deployments lag because they must reconcile the computational and organizational complexity of fusing diverse asset twins, legacy GIS layers and live IoT streams. Bridging this scale gap will therefore require (1) hierarchical data ontologies that allow semantic inheritance from component to district, (2) edge-cloud federations capable of streaming analytics without prohibitive bandwidth, and (3) governance frameworks that align municipal, utility, and private-sector stakeholders around shared value propositions. Developing such multi-scale, interoperable architectures is essential if DTs are to evolve from isolated “one-off” pilots into a cohesive digital fabric underpinning resilient, resource-efficient cities of the future.

8. Discussion, challenges, and future work

Fig. 9 show the distribution of papers between 2020 and 2024. The

dotted line shows that this survey was taken in July 2024, and that the calendar year was incomplete. The results reflect this, showing a general increase in submissions for most categories that follow trends shown between 2020 and 2023. It illustrates the overall distribution of studies across analytical dimensions, highlighting the interconnections among research types, methods, lifecycle phases, and sectors. The figure emphasizes that technical studies dominate the field, particularly within operation and maintenance phases, while theoretical works remain comparatively limited. It also shows that most DT applications are concentrated in the building and infrastructure sectors, with fewer examples addressing city-scale or cross-sectoral implementations. This synthesis underlines the field's practical maturity but conceptual fragmentation, suggesting the need for broader integration across scales and lifecycle stages.

Fig. 10 illustrates the relationships between various research papers and their corresponding research categories or themes. Each horizontal flow represents a connection between a specific reference and one or more categories it contributes to. The width of each flow indicates the strength or frequency of the association. The most represented categories are Operation and Maintenance, Asset, Technical, Monitoring, and Buildings. Less frequently addressed categories include Roads, Tunnels, and Bridges. This type of diagram highlights dominant themes in the literature and helps identify potential publication biases or gaps in coverage, such as the underrepresentation of certain infrastructure types.

A synthesis across the reviewed sectors reveals that while DT adoption pathways differ in maturity and focus, they converge on a shared set of technical and organizational challenges. In safety-critical domains such as bridges and tunnels, sensor-driven and simulation-based twins demonstrate strong predictive capabilities but remain constrained by calibration demands and data fragmentation. Conversely, building and construction management sectors benefit most from modelling-centred twins that leverage BIM, ontology mapping, and AI for operational efficiency and sustainability benchmarking, yet they struggle with interoperability and version control. Urban and road infrastructure applications show the most promise for integrating edge-AI, GIS, and multi-criteria decision frameworks, but suffer from scalability and

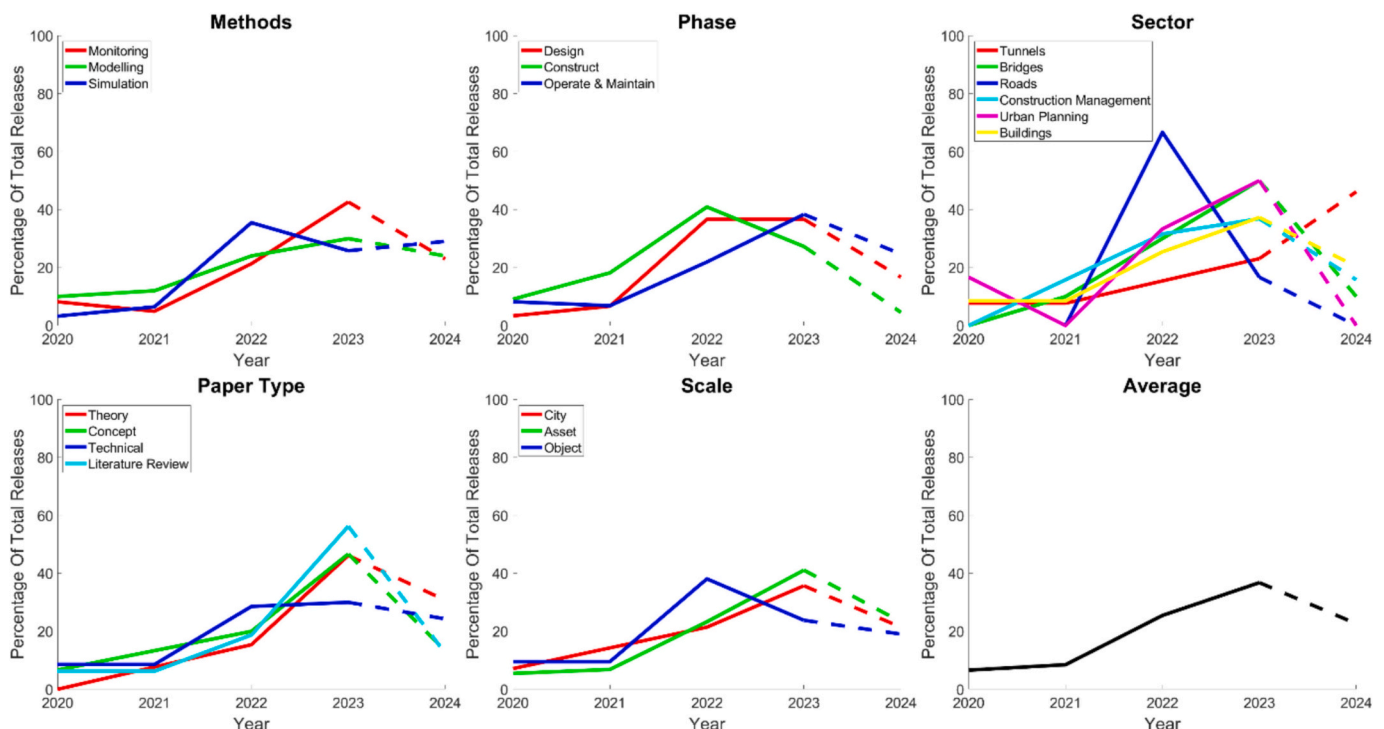


Fig. 9. Yearly trends in DT research publications from 2020 to mid-2024.

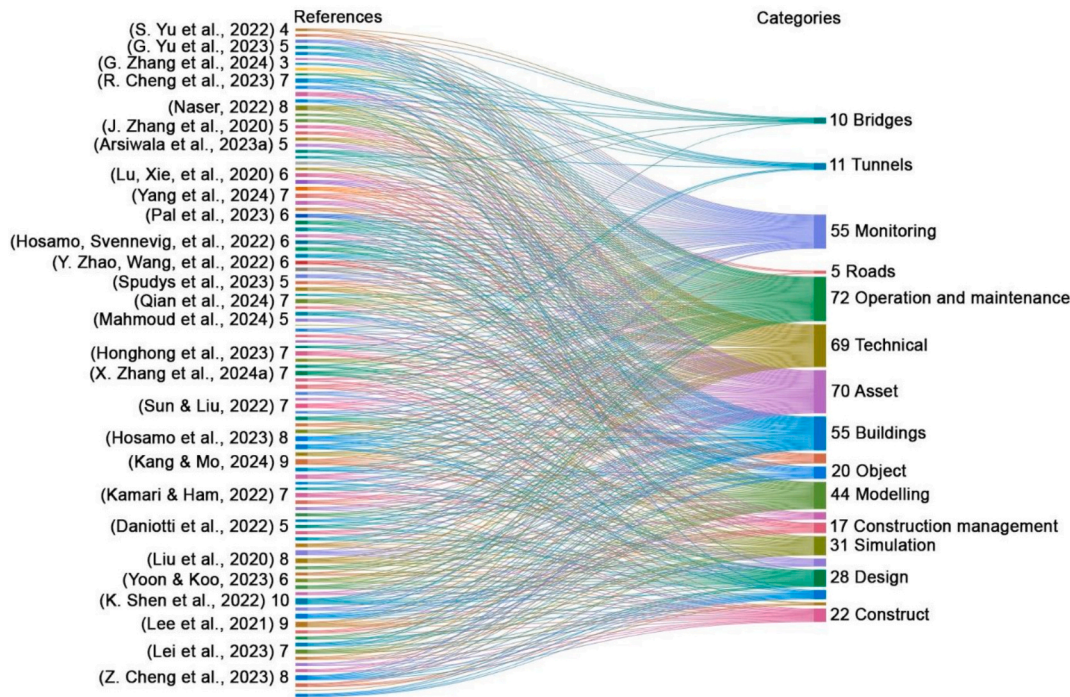


Fig. 10. Sankey diagram of the distribution of references across research categories.

governance limitations. These findings indicate that methodological strengths are context-dependent: monitoring works best for dynamic systems requiring continuous feedback, modelling for semantically rich environments, and simulation for predictive policy or performance assessment. Future research should therefore pursue (1) hybrid pipelines that dynamically switch between these methods depending on data quality and lifecycle stage, (2) harmonised data ontologies and open-source middleware to support cross-sector integration, and (3) benchmarking protocols that quantify transferability of DT frameworks between domains. The next generation of DTs can evolve from sector-specific prototypes into a cohesive digital infrastructure ecosystem by embedding these strategies within interoperable, ethically governed, and scalable architectures.

8.1. Discussion

DT research in the construction and built environment context has demonstrated remarkable versatility and potential. The integration of technologies such as BIM, IoT sensing, and data analytics highlights how real-time monitoring, modelling, and simulation can contribute to more informed decision-making across a diverse range of infrastructure projects. Through these three foundational methods, scholars have developed robust frameworks that not only capture the current conditions of physical assets but also predict future behaviours, enabling proactive interventions for construction, operation, and maintenance. This bridging of physical and virtual realms underscores the interdisciplinary character of DT research, where insights from civil engineering, computer science, and data analytics coalesce to address challenges that transcend traditional sector boundaries.

Despite these demonstrated benefits, the studies surveyed reveal that DT adoption remains uneven and fragmented, often limited by issues of interoperability and standardization. The exchange of data between proprietary platforms and technologies can be cumbersome, requiring significant effort to reconcile differences in data formats and classification systems. As a result, many DT deployments are siloed, confining their impact to discrete project phases (e.g., design or maintenance) rather than spanning the full lifecycle of an infrastructure asset. The

difficulties in achieving cross-platform integration are exacerbated by high upfront costs associated with sensor installation and cloud or edge computing infrastructure. These financial barriers can be especially prohibitive for smaller organizations and municipalities, which often lack the budgetary flexibility to trial emerging technologies at scale.

At the same time, the growing body of research on AI-driven analytics and federated learning approaches points to a future where some of these challenges may be addressed. Fig. 11 presents the challenges and future directions [32]. The themes were derived by aggregating recurrent issues and research priorities discussed in Sections 3 to 7, representing both empirical findings and conceptual insights on DT implementation barriers and opportunities [32]. Stakeholders could better harness DTs in real-time without excessive reliance on high-bandwidth networks or centralized computing clusters by pushing computation closer to the data source and distributing data-processing tasks. This direction aligns with an increasingly urgent need for rapid responsiveness in infrastructure systems, particularly when dealing with complex phenomena such as earthquakes, flooding, or severe traffic congestion [106]. In addition to building performance optimization, issues of governance, privacy, and ethical AI use remain pressing topics within the DT domain. Scholars have emphasized that more concerted efforts toward policy frameworks and stakeholder collaboration are necessary for scaling DT implementations responsibly [4].

Research gaps persist where DT concepts could have significant impact, such as automating building permitting processes, standardizing occupant-centric performance metrics, and addressing the intricacies of retrofitting historical structures under evolving regulatory constraints. Exploring these less-studied areas would enrich the current state of DT literature, opening avenues for more holistic, end-to-end management of infrastructure throughout its entire lifecycle. Future investigations can further harness DTs potential to facilitate not just greater operational efficiency and sustainability, but also equitable development, community resilience, and long-term societal benefit by addressing the multi-faceted technical, financial, and governance-related barriers.

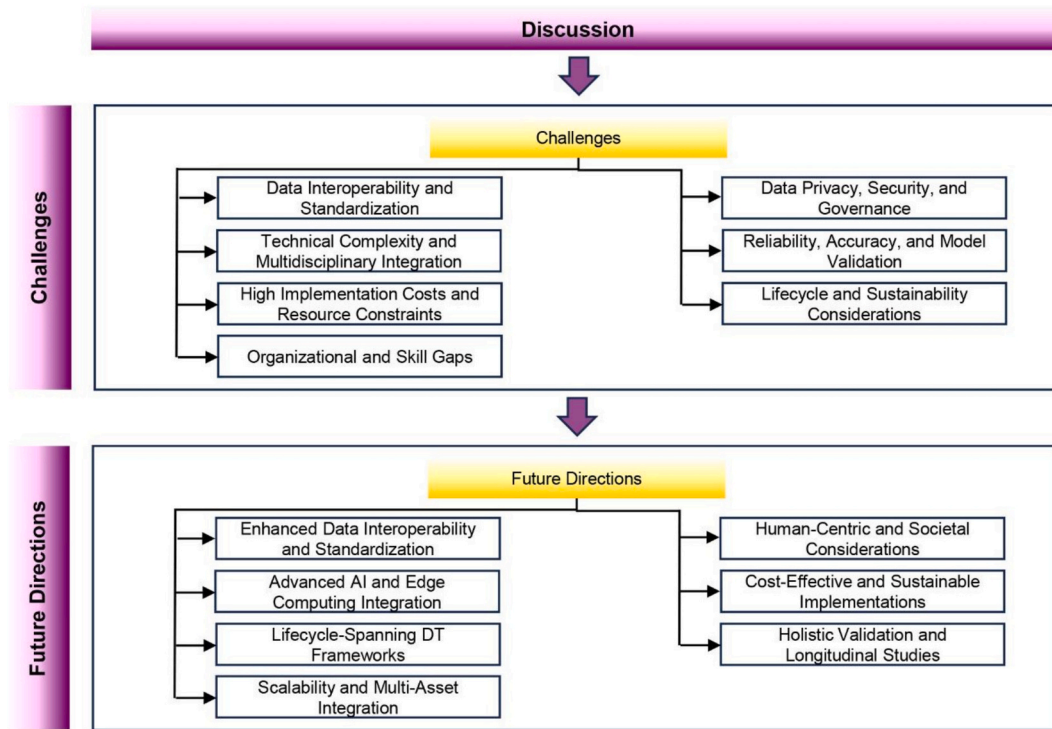


Fig. 11. Challenges and future directions of DTs.

8.2. Challenges

Despite the growing body of literature illustrating the advantages of DT technologies in the infrastructure and built environment, several persistent challenges hinder their broader adoption and full potential. These challenges are multidimensional, spanning technical, organizational, and societal barriers:

- 1. Data Interoperability and Standardization:** (1) *Heterogeneous Data Sources:* Projects commonly involve diverse file formats, sensor outputs, and BIM standards, leading to fragmentation and difficulties in integrating data into cohesive DT frameworks; (2) *Lack of Unified Protocols:* Inconsistent ontologies and taxonomies reduce the ability to share DT data between stakeholders, slowing collaborative research and practical implementations; (3) *Semantic Enrichment Gaps:* While IFC and other standards exist, many do not accommodate emerging DT requirements such as edge computing, advanced analytics, or domain-specific semantics.
- 2. Technical Complexity and Multidisciplinary Integration:** (1) *Sensor and Hardware Limitations:* Sensor calibration, reliability under varying environmental conditions, and real-time data transmission remain non-trivial challenges, particularly for large-scale or harsh infrastructural environments; (2) *Complex System Architectures:* DTs typically combine BIM, AI, IoT, and physics-based simulations, each requiring specialized expertise. Many practitioners find it challenging to develop and maintain robust systems that integrate all these components effectively; (3) *Scalability and Real-Time Analytics:* As infrastructures expand city-wide or region-wide, the sheer volume of incoming data strains existing computing, storage, and network capacities. Synchronizing and processing data in near real-time often requires advanced techniques such as edge computing or federated learning.
- 3. High Implementation Costs and Resource Constraints:** (1) *Financial Barriers:* Installing sensors, commissioning cloud or edge computing infrastructures, and training personnel represent substantial initial investments. Smaller organizations and local

governments may struggle to justify these expenditures without guaranteed returns; (2) *Long-Term Maintenance:* Sustaining a DT over an asset's lifecycle calls for periodic sensor updates, software upgrades, and continuous data validation, factors that raise both technical and financial burdens.

- 4. Organizational and Skill Gaps:** (1) *Limited Skilled Workforce:* DTs demand knowledge in data analytics, software engineering, AI, and domain-specific engineering. Many construction and facility management teams lack this multidisciplinary skill set, delaying DT adoption; (2) *Siloed Project Phases:* Traditional design, construction, and operation workflows are often siloed, making it difficult to adopt a shared DT that spans the entire infrastructure lifecycle; (3) *Resistance to Change:* Shifting established processes to data-driven, real-time systems can encounter organizational inertia. Cultural barriers and unclear return on investment can exacerbate resistance among key stakeholders.
- 5. Data Privacy, Security, and Governance:** (1) *Sensitive Information Exposure:* Real-time monitoring in buildings and city-scale DTs may capture occupant data and personal information, raising privacy concerns; (2) *Cybersecurity Threats:* Integrating IoT devices, cloud services, and edge computing creates numerous points of vulnerability. Malicious actors could disrupt sensor streams, manipulate control systems, or compromise sensitive design data; (3) *Regulatory and Ethical Uncertainty:* Existing policies and laws often lag rapid technological developments. Clear guidelines on data ownership, usage rights, and ethical AI applications are still evolving.
- 6. Reliability, Accuracy, and Model Validation:** (1) *Model Drift and Calibration:* Ensuring that virtual models accurately reflect as-built conditions requires continuous calibration and data validation. When sensor inputs degrade or fail, the DT can diverge from the physical asset; (2) *Predictive Model Limitations:* ML and AI-based predictive models can be prone to false positives or negatives if training datasets are incomplete or unrepresentative. Model interpretability also remains a challenge in safety-critical applications.
- 7. Lifecycle and Sustainability Considerations:** (1) *Fragmented Lifecycle Integration:* While DTs hold promise for end-to-end asset

management, implementing them consistently, across design, construction, and operation and maintenance, remains complex; (2) *Environmental Footprint*: DTs require considerable computational power, sensor hardware, and network infrastructure. Balancing these needs with sustainability goals calls for energy-efficient computing solutions and hardware reuse strategies.

In addressing these challenges, future research and industry efforts must concentrate on creating robust interoperability frameworks, developing clear cost-benefit analyses, and establishing strong stakeholder collaboration to foster trust and collective ownership. More sophisticated AI-driven analytics, as well as policy and governance structures tailored to data privacy and security, will also be key to ensuring that DT technology matures into a trusted and indispensable asset for infrastructure and the built environment.

8.3. Future directions

While the promise of DT technologies in infrastructure and the built environment is increasingly evident, significant opportunities remain to refine their development, deployment, and impact. Future research can benefit from targeted explorations in several key areas, as shown in Fig. 12. Higher radial values indicate themes emphasized most strongly in the *Future work* discussion.

- Enhanced Data Interoperability and Standardization:** (1) Robust Data Schemas and Ontologies: Developing universal standards for merging BIM, IoT sensor data, GIS, and AI-driven insights will help unify diverse data sources. Such schemas must accommodate high-dimensional, domain-specific data while allowing flexible extensions for emerging technologies; (2) Inter-Platform Communication Protocols: Future studies can focus on creating lightweight, open-source interfaces or middleware solutions that streamline communication among different BIM software, databases, and real-time analytics engines.
- Advanced AI and Edge Computing Integration:** (1) Real-Time Predictive Analytics: There is a growing need to design resource-efficient ML models capable of handling vast, real-time data streams. Tools such as federated learning, distributed training, and on-device (edge) inference can significantly reduce latency and bandwidth demands; (2) Explainable AI (XAI): As DT systems increasingly incorporate AI for critical decision-making (e.g., structural health monitoring, evacuation planning), improving the interpretability and trustworthiness of these algorithms will be crucial for industry adoption and regulatory compliance.

- Lifecycle-Spanning DT Frameworks:** (1) Seamless Phase Transitions: Although DTs can theoretically support design, construction, and operation, few frameworks manage smooth data handoffs across all phases. Future research must detail how data models and analytics can adapt as projects shift from planning to active construction and eventual facility management; (2) Retrofitting and Renovation Support: Existing studies largely focus on new assets; investigations of how to embed DTs into older or historically significant structures would enable cost-efficient, minimally intrusive lifecycle extensions for aging infrastructure.
- Scalability and Multi-Asset Integration:** (1) City-Wide Deployments: As urban environments grow in complexity, future work should examine how to create digital replicas of entire neighbourhoods or cities to inform large-scale planning, climate resilience strategies, and resource sharing; (2) Multi-Vector Simulations: Research into coupling energy, transportation, water, and communication networks within integrated DTs can unveil interdependencies and optimize resource usage under diverse operational conditions.
- Human-Centric and Societal Considerations:** (1) User Engagement and Accessibility: Many DT projects remain developer- or engineer-oriented. Incorporating user-friendly interfaces and visualizations will empower non-technical stakeholders—such as city planners, building occupants, and community members—to benefit from DT insights; (2) Ethical and Privacy Frameworks: As DTs increasingly capture occupant data and real-time activity, interdisciplinary studies on data governance, privacy, cybersecurity, and regulatory compliance will be essential. Empirical evaluations of social acceptance and public trust can guide balanced policies that protect user interests without inhibiting innovation.
- Cost-Effective and Sustainable Implementations:** (1) Economic Viability: Future studies should prioritise creating transparent return-on-investment (ROI) models that quantify potential savings in operations, maintenance, and risk mitigation. Demonstrating tangible benefits could inspire stronger adoption among budget-constrained organizations; (2) Low-Carbon DT Ecosystems: Since DT solutions may consume significant computational power, exploring green data centres, energy-efficient hardware, and optimized communication protocols can help align DT adoption with broader sustainability goals.
- Holistic Validation and Longitudinal Studies:** (1) Performance Benchmarks and Metrics: Standardized benchmarks will enable fair comparisons among different DT solutions, supporting more objective decisions on technology investments; (2) Long-Term Field Evaluations: Comprehensive case studies observing DT performance over multiple years, from design through post-construction, can offer richer insights on reliability, adaptability, and cost-effectiveness across an asset's lifespan.

Additionally, future research should move beyond generic calls for interoperability and data integration by engaging with specific technical directions and standards emerging from both academia and industry. Promising pathways include (1) the adoption of IFC 4.3 and the International Organization for Standardization (ISO) 19,650 framework to ensure standardized data exchange and lifecycle information management; (2) the development of Application Programming Interface (API)-driven middleware and open-source connectors for synchronizing DT data with BIM and IoT environments; (3) the integration of edge-cloud hybrid computing and federated learning architectures to improve privacy-preserving, decentralized analytics; and (4) the establishment of semantic ontologies such as Building-Related Information and Cyber-Knowledge (BRICK), Building Topology Ontology (BOT), and DTs Definition Language (DTD) to formalize asset-level knowledge representation across platforms.

Building on the reviewed evidence, research should focus on benchmark datasets for DT validation, performance metrics for real-time

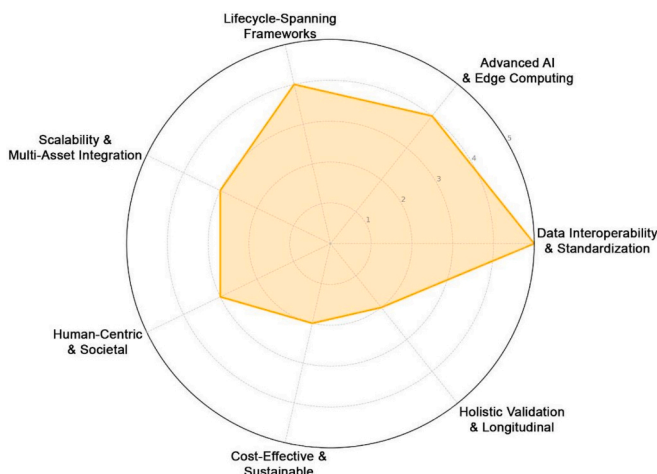


Fig. 12. Future DT research priorities.

synchronization, and domain-specific reference frameworks such as the Centre for Digital Built Britain (CDBB)'s Gemini Principles and the DT Hub Maturity Model (DTH-MM) to harmonize cross-sector implementations. Empirical studies could also evaluate how DT-driven lifecycle feedback loops, linking design, construction, and operation data, improve predictive accuracy and sustainability outcomes at both the asset and city scales. These directions collectively point toward a more rigorous, standard-compliant, and data-centric ecosystem for DTs in infrastructure, capable of scaling from experimental prototypes to institutionalized digital infrastructure intelligence systems.

The proposed future directions are closely aligned with the challenges identified in the preceding section. Specifically, efforts to enhance data interoperability and standardization directly respond to persistent issues of fragmented data sources and inconsistent ontologies. Advancements in AI and edge computing integration address the technical complexity and scalability barriers that currently hinder real-time DT deployment. The development of lifecycle-spanning frameworks and multi-asset integration corresponds to the identified organizational and skill gaps and the difficulty of achieving seamless data continuity across project phases. Moreover, calls for human-centric, ethical, and sustainable DT ecosystems stem from challenges surrounding data privacy, governance, and environmental impact. Explicitly linking these challenges to their corresponding research trajectories highlights that overcoming them will serve as the foundation for establishing robust, trustworthy, and widely adoptable DT ecosystems in the built environment.

Future DT research can move beyond isolated demonstrations toward comprehensive, reliable, and ethically governed ecosystems by delving deeper into these directions. This evolution will better position DTs to address increasingly complex societal demands, enhancing not only the technical integrity of infrastructure projects but also fostering resilience, sustainability, and stakeholder collaboration at all scales.

9. Conclusion

This paper provided a review of how DTs have been developed and applied within infrastructure and the built environment. It offered an integrated overview of the research landscape, examining the types of methods used, the sectors and lifecycle phases addressed, and the scales of implementation. The findings revealed that most DT studies combine real-time monitoring, semantic modelling, and physics- or data-driven simulation to ensure synchronization between physical assets and their virtual counterparts. While each approach has its own value, hybrid workflows have emerged as the most effective, offering both explanatory depth and temporal fidelity.

The review highlighted that DT adoption has been most prominent in building-related applications, followed by construction management and specialized infrastructure such as tunnels, bridges, and roads. In safety-critical domains, researchers have prioritized high-resolution sensing and finite-element modelling, whereas energy-intensive facilities have focused on the integration of IoT and AI to support operational optimisation. A significant trend was the concentration of DT deployments in the operation and maintenance phase of the asset lifecycle, where the benefits are most immediate and tangible. However, far fewer studies explored DT implementation during the design or construction phases, resulting in fragmented information flow and the loss of potentially valuable data during project hand-offs. Moreover, the review found that asset-level DTs dominate the field; object-scale studies remain isolated, and city-scale pilots are still in their infancy due to persistent interoperability and governance challenges.

Despite the rapid progress observed in recent years, several challenges continue to hinder the widespread adoption of DTs. These include heterogeneous data formats, high initial investment costs, a shortage of cross-disciplinary expertise, and unresolved issues surrounding privacy, security, and ethics. Such barriers often restrict DT deployments to single phases, individual organizations, or proprietary technology

ecosystems, limiting their potential as integrated decision-support systems.

Looking ahead, future research must prioritise the development of robust interoperability frameworks, resource-efficient artificial intelligence and edge-computing solutions, and lifecycle-spanning data pipelines that preserve information continuity from design through to decommissioning. Advancing DTs to multi-asset and city-scale applications will also require stronger governance structures, open standards, and clear evidence of long-term value to justify investment. Crucially, the successful realisation of these goals will depend on close collaboration between engineers, computer scientists, policy-makers, and end-users.

This review found that (RQ1) DTs have been applied most extensively in building and operation and maintenance phases, (RQ2) key enabling methods include monitoring, modelling, and simulation integrated via BIM, IoT, and AI, and (RQ3) major challenges remain in data interoperability, lifecycle integration, and scalability. These insights highlight future research priorities for developing interoperable, cross-sector DT frameworks.

If such coordinated efforts are achieved, DTs have the potential to evolve from isolated proofs of concept into an interconnected digital ecosystem that underpins resilient, low-carbon, and adaptable infrastructure. In doing so, they could fulfil their promise as indispensable decision-support tools for shaping the sustainable built environment of the future.

CRedit authorship contribution statement

Soheila Kookalani: Writing – review & editing, Writing – original draft, Visualization, Methodology. **Stephen Green:** Writing – original draft, Visualization. **Peihang Luo:** Writing – original draft, Visualization. **Hamidreza Alavi:** Writing – original draft. **Erika Parn:** Writing – original draft. **Zhaojie Sun:** Writing – original draft. **Ioannis Brilakis:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors gratefully acknowledge Judith Fauth for her contribution to the development of this paper.

Data availability

No data was used for the research described in the article.

References

- [1] Y. Shen, J. Ling, X. Li, H. Li, S. Feng, H. Zhu, Holistic digital-twin-based framework to improve tunnel lighting environment: from methodology to application, *Build. Environ.* 224 (2022) 109562, <https://doi.org/10.1016/j.buildenv.2022.109562>.
- [2] S. Kookalani, E. Parn, I. Brilakis, Interpretable machine learning approaches for assessing maximum force in fiber-reinforced composites, in: *Proceedings of the International Symposium on Automation and Robotics in Construction*, 2024, pp. 645–652, <https://doi.org/10.22260/ISARC2024/0084>.
- [3] S. Kookalani, E. Parn, I. Brilakis, P. Luo, H. Alavi, M. Wang, S. Green, Matching reused elements to structural designs for steel reuse, in: *ASCE International Conference on Computing in Civil Engineering*, 2024, <https://doi.org/10.17863/CAM.108198>.
- [4] S. Yoon, J. Koo, In situ model fusion for building digital twinning, *Build. Environ.* 243 (2023), <https://doi.org/10.1016/j.buildenv.2023.110652>.
- [5] H. Alavi, M. Wang, G. Wang, S. Kookalani, A. Mathew, I. Brilakis, Digital twins in construction: leveraging point cloud data and BIM for monitoring project schedule, in: *ASCE International Conference on Computing in Civil Engineering*, 2024, <https://doi.org/10.17863/CAM.107992>.

- [6] H. Alavi, W. Chen, M. Wang, S. Kookalani, I. Brilakis, Enhancing building maintenance efficiency through BIM-LiDAR integration, in: *Proceedings of the European Conference on Computing in Construction 5*, 2024 doi:10.35490/EC3.2024.291.
- [7] M. Wang, Y. Wen, Y. Pan, H. Alavi, S. Kookalani, W. Chen, I. Brilakis, Automatic extraction of building features for building facades based on laser scanning technology, in: *Proceedings of the European Conference on Computing in Construction 2024*, 2024, pp. 519–525, <https://doi.org/10.35490/EC3.2024.234>.
- [8] S. Kookalani, H.H. Aung, GFRP elastic Gridshell structures: a review of methods, research, applications, opportunities, and challenges, *J. Civil Eng. Mater. Appl.* 7 (2023) 71–94, <https://doi.org/10.22034/JCEMA.2023.401834.1111>.
- [9] X. Li, L. Tang, J. Ling, C. Chen, Y. Shen, H. Zhu, Digital-twin-enabled JIT design of rock tunnel: methodology and application, *Tunn. Undergr. Space Technol.* 140 (2023), <https://doi.org/10.1016/j.tust.2023.105307>.
- [10] G. Yu, Y. Wang, Z. Mao, M. Hu, V. Sugumaran, Y.K. Wang, A digital twin-based decision analysis framework for operation and maintenance of tunnels, *Tunn. Undergr. Space Technol.* 116 (2021), <https://doi.org/10.1016/j.tust.2021.104125>.
- [11] Y. Gao, H. Li, G. Xiong, H. Song, AIoT-informed digital twin communication for bridge maintenance, *Autom. Constr.* 150 (2023), <https://doi.org/10.1016/j.autcon.2023.104835>.
- [12] Y. Yang, Y. Zhu, C.S. Cai, Research progress and prospect of digital twin in bridge engineering, *Adv. Struct. Eng.* 27 (2024) 333–352, <https://doi.org/10.1177/13694332231218764>.
- [13] Y. Peng, M. Zhang, F. Yu, J. Xu, S. Gao, Digital twin hospital buildings: an exemplary case study through continuous lifecycle integration, *Adv. Civil Eng.* 2020 (2020), <https://doi.org/10.1155/2020/8846667>.
- [14] B. Daniotti, G. Masera, C.M. Bolognesi, S.L. Spagnolo, A. Pavan, G. Iannaccone, M. Signorini, S. Ciuffreda, C. Mirarchi, M. Lucky, M. Cucuzza, The development of a BIM-based interoperable toolkit for efficient renovation in buildings: from BIM to digital twin, *Buildings* 12 (2022), <https://doi.org/10.3390/buildings12020231>.
- [15] T. Wang, V.J.L. Gan, D. Hu, H. Liu, Digital twin-enabled built environment sensing and monitoring through semantic enrichment of BIM with SensorML, *Autom. Constr.* 144 (2022), <https://doi.org/10.1016/j.autcon.2022.104625>.
- [16] S. Kookalani, B. Cheng, Structural analysis of GFRP elastic Gridshell structures by particle swarm optimization and least square support vector machine algorithms, *J. Civil Eng. Mater. Appl.* 5 (2021) 139–150, <https://doi.org/10.22034/JCEMA.2021.304981.1064>.
- [17] S. Xiang, B. Cheng, S. Kookalani, An analytic solution for form finding of GFRP elastic gridshells during lifting construction, *Compos. Struct.* 244 (2020), <https://doi.org/10.1016/j.compstruct.2020.112290>.
- [18] S. Kookalani, B. Cheng, S. Xiang, Shape optimization of GFRP elastic gridshells by the weighted Lagrange ϵ -twin support vector machine and multi-objective particle swarm optimization algorithm considering structural weight, *Structures* 33 (2021) 2066–2084, <https://doi.org/10.1016/j.istruc.2021.05.077>.
- [19] S. Xiang, B. Cheng, L. Zou, S. Kookalani, An integrated approach of form finding and construction simulation for glass fiber-reinforced polymer elastic gridshells, *Struct. Des. Tall Spec.* 29 (2020), <https://doi.org/10.1002/TAL.1698>.
- [20] S. Xiang, B. Cheng, S. Kookalani, J. Zhao, An analytic approach to predict the shape and internal forces of barrel vault elastic gridshells during lifting construction, *Structures* 29 (2021) 628–637, <https://doi.org/10.1016/j.istruc.2020.11.032>.
- [21] S. Kookalani, B. Cheng, J.L.C. Torres, Structural performance assessment of GFRP elastic gridshells by machine learning interpretability methods, *Front. Struct. Civ. Eng.* 16 (2022) 1249–1266, <https://doi.org/10.1007/s11709-022-0858-5>.
- [22] S. Kookalani, E. Parn, I. Brilakis, S. Dirar, M. Theofanous, A. Faramarzi, M. A. Mahdavi, Q. Feng, Trajectory of building and structural design automation from generative design towards the integration of deep generative models and optimization: a review, *J. Build. Eng.* 97 (2024) 110972, <https://doi.org/10.1016/j.jobe.2024.110972>.
- [23] S. Kookalani, D. Shen, Leo, L. Zhu, M. Lindsey, An overview of optimal damper placement methods in structures, *Trans. Civ. Eng.* 46 (2022) 1785–1804, <https://doi.org/10.1007/s40996-021-00752-2>.
- [24] S. Kookalani, S. Nyunn, S. Xiang, Form-finding of lifting self-forming GFRP elastic gridshells based on machine learning interpretability methods, *Struct. Eng. Mech.* 84 (2022) 605–618, <https://doi.org/10.12989/sem.2022.84.5.605>.
- [25] S. Kookalani, B. Cheng, Parametric-insensitive nonparallel support vector regression for structural stress prediction of GFRP elastic gridshell structures, in: *International Conference on New Research and Achievements in Science, Engineering and Technologies*, Seoul, South Korea, 2021. <https://www.isnac.ir/XDKF-GDBEG> (accessed June 11, 2023).
- [26] H. Alavi, S. Kookalani, F. Rahimian, N. Forcada, Introduction of methodology for BIM & DSS, *Integr. Build. Intellig.* (2024) 31–42, https://doi.org/10.1007/978-3-031-68865-2_3.
- [27] H. Alavi, S. Kookalani, F. Rahimian, N. Forcada, BIM-based DSS for building condition assessment, *Integr. Build. Intellig.* (2024) 59–78, https://doi.org/10.1007/978-3-031-68865-2_5.
- [28] H. Alavi, S. Kookalani, F. Rahimian, N. Forcada, BIM-based DSS for HVAC root-cause detection, *Integr. Build. Intellig.* (2024) 43–57, https://doi.org/10.1007/978-3-031-68865-2_4.
- [29] G. Editors, K. Ruikar, K. Kotecha, S. Sandbhor, A. Thomas, M. Deng, C. C. Menassa, A. Professor, V.R. Kamat, From BIM to digital twins: a systematic review of the evolution of intelligent building representations in the AEC-FM industry, *ITcon* 26 (2021) 58–83, <https://doi.org/10.36680/J.ITCON.2021.005>.
- Special Issue Next Generation ICT - How Distant Is Ubiquitous Computing?, Pg. 58–83.
- [30] H. Liu, H. Su, S. Kookalani, M. Cao, Y. Lei, W. Wu, Attention-enhanced spatiotemporal deep learning-based automatic warning model with uncertainty estimation in dam safety, *Struct. Health Monit.* (2025), <https://doi.org/10.1177/14759217251358537>.
- [31] S. Kookalani, H. Alavi, F.P. Rahimian, Structural Design and Optimization of Lifting Self-forming GFRP Elastic Gridshells based on Machine Learning, Routledge, 2025, <https://doi.org/10.1201/9781003565055>.
- [32] C. Boje, A. Guerriero, S. Kubicki, Y. Rezgui, Towards a semantic construction digital twin: directions for future research, *Autom. Constr.* 114 (2020) 103179, <https://doi.org/10.1016/J.AUTCON.2020.103179>.
- [33] M.A. Mahdavi, A. Faramarzi, S. Dirar, M. Theofanous, S. Jowhari Moghadam, Q. Feng, S. Kookalani, E. Parn, I. Brilakis, Optimised steel frame design using reclaimed steel: logistics impact on reuse efficiency, *J. Constr. Steel Res.* 236 (2026) 110046, <https://doi.org/10.1016/J.JCSR.2025.110046>.
- [34] P. Luo, E. Parn, S. Kookalani, I. Brilakis, TailorAlert: large language model-based personalized alert generation system for road infrastructure management with digital twins, in: *ASCE International Conference on Computing in Civil Engineering*, 2024, <https://doi.org/10.17863/CAM.109005>.
- [35] W. Albalkhy, D. Karmaoui, L. Ducoulombier, Z. Lafhaj, T. Linner, Digital twins in the built environment: definition, applications, and challenges, *Autom. Constr.* 162 (2024) 105368, <https://doi.org/10.1016/J.AUTCON.2024.105368>.
- [36] W. Hu, W. Wang, C. Ai, J. Wang, W. Wang, X. Meng, J. Liu, H. Tao, S. Qiu, Machine vision-based surface crack analysis for transportation infrastructure, *Autom. Constr.* 132 (2021) 103973, <https://doi.org/10.1016/J.AUTCON.2021.103973>.
- [37] S. Kookalani, E. Parn, I. Brilakis, Assessment of the bearing capacity of double shear-bolted connections in structural steel, in: *31st EG-ICE International Workshop on Intelligent Computing in Engineering*, 2024, <https://doi.org/10.17863/CAM.108741>.
- [38] S. Kookalani, I. Brilakis, G.B. Ozturk, K. Oti-Sarpong, Machine learning-driven structural analysis of lifting self-forming GFRP elastic gridshells, in: *The 8th International Project and Construction Management Conference, IPCMC 2024*, 2024, <https://doi.org/10.17863/CAM.108491>.
- [39] G.B. Ozturk, I. Brilakis, S. Kookalani, Generative AI research fields in the AEC-FM industry, in: *The 8th International Project and Construction Management Conference, IPCMC*, 2024, <https://doi.org/10.17863/CAM.108492>.
- [40] X. Wang, Y. Zhang, H. Li, C. Wang, P. Feng, Applications and challenges of digital twin intelligent sensing technologies for asphalt pavements, *Autom. Constr.* 164 (2024) 105480, <https://doi.org/10.1016/J.AUTCON.2024.105480>.
- [41] Z. Liu, Y. Mo, B. Hou, M. Li, W. Li, C. Xu, Digital twin modelling approaches and applications in urban infrastructure operations and maintenance, *Autom. Constr.* 179 (2025) 106445, <https://doi.org/10.1016/J.AUTCON.2025.106445>.
- [42] H. Gong, D. Su, S. Zeng, X. Chen, Advancements in digital twin modeling for underground spaces and lightweight geometric modeling technologies, *Autom. Constr.* 165 (2024) 105578, <https://doi.org/10.1016/J.AUTCON.2024.105578>.
- [43] W. Saif, S.R. RazaviAlavi, M. Kassem, Construction digital twin: a taxonomy and analysis of the application-technology-data triad, *Autom. Constr.* 167 (2024) 105715, <https://doi.org/10.1016/J.AUTCON.2024.105715>.
- [44] L. Lian, Y. Yan, Digital visual sensing design teaching using digital twins, *Adv. Civil Eng.* 2022 (2022), <https://doi.org/10.1155/2022/9311246>.
- [45] G. Zhang, Z. Zhou, C. Wan, Z. Ding, Z. Wu, L. Xie, S. Xue, Substructural damage identification in a digital twin framework using heterogeneous response reconstruction, *Adv. Struct. Eng.* (2024), <https://doi.org/10.1177/13694332241242984>.
- [46] I. Esmaili, D. Simeone, A general contractor's perspective on construction digital twin: implementation, impacts and challenges, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13040978>.
- [47] S. Honghong, Y. Gang, L. Haijiang, Z. Tian, J. Annan, Digital twin enhanced BIM to shape full life cycle digital transformation for bridge engineering, *Autom. Constr.* 147 (2023) 104736, <https://doi.org/10.1016/j.autcon.2022.104736>.
- [48] Q. Lu, L. Chen, S. Li, M. Pitt, Semi-automatic geometric digital twinning for existing buildings based on images and CAD drawings, *Autom. Constr.* 115 (2020), <https://doi.org/10.1016/j.autcon.2020.103183>.
- [49] M. Shahinmoghadam, W. Natephra, A. Motamedi, BIM- and IoT-based virtual reality tool for real-time thermal comfort assessment in building enclosures, *Build. Environ.* 199 (2021), <https://doi.org/10.1016/j.buildenv.2021.107905>.
- [50] A. Agrawal, R. Thiel, P. Jain, V. Singh, M. Fischer, Digital twin: Where do humans fit in? *Autom. Constr.* 148 (2023) <https://doi.org/10.1016/j.autcon.2023.104749>.
- [51] D.G.J. Opoku, S. Perera, R. Osei-Kyei, M. Rashidi, K. Bamdad, T. Famakinwa, Digital twin for indoor condition monitoring in living labs: university library case study, *Autom. Constr.* 157 (2024), <https://doi.org/10.1016/j.autcon.2023.105188>.
- [52] M.Z. Naser, Digital twin for next gen concretes: on-demand tuning of vulnerable mixtures through explainable and anomalous machine learning, *Cem. Concr. Compos.* 132 (2022), <https://doi.org/10.1016/j.cemconcomp.2022.104640>.
- [53] H. Sun, Z. Liu, Research on intelligent dispatching system management platform for construction projects based on digital twin and BIM technology, *Adv. Civil Eng.* 2022 (2022), <https://doi.org/10.1155/2022/8273451>.
- [54] Z. Liu, A. Jiang, A. Zhang, Z. Xing, X. Du, Intelligent prediction method for operation and maintenance safety of prestressed steel structure based on digital twin technology, *Adv. Civil Eng.* 2021 (2021), <https://doi.org/10.1155/2021/6640198>.

- [55] R. Cheng, L. Hou, S. Xu, A review of digital twin applications in civil and infrastructure emergency management, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13051143>.
- [56] F. Hodavand, I.J. Ramaji, N. Sadeghi, Digital twin for fault detection and diagnosis of building operations: a systematic review, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13061426>.
- [57] B. Lei, P. Janssen, J. Stoter, F. Biljecki, Challenges of urban digital twins: a systematic review and a Delphi expert survey, *Autom. Constr.* 147 (2023), <https://doi.org/10.1016/j.autcon.2022.104716>.
- [58] G.B. Ozturk, Digital twin research in the AECO-FM industry, *J. Build. Eng.* 40 (2021), <https://doi.org/10.1016/j.jobte.2021.102730>.
- [59] A.A. Alnaser, A. Hassan Ali, H.H. Elmousalami, A. Elyamany, A. Gouda Mohamed, Assessment framework for BIM-digital twin readiness in the construction industry, *Buildings* 14 (2024), <https://doi.org/10.3390/buildings14010268>.
- [60] G. Desogus, C. Frau, E. Quaquero, G. Rubiu, From building information model to digital twin: a framework for building thermal comfort monitoring, visualizing, and assessment, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13081971>.
- [61] P. Spudys, N. Afxentiou, P.Z. Georgali, E. Klumbyte, A. Jurelionis, P. Fokaides, Classifying the operational energy performance of buildings with the use of digital twins, *Energ. Buildings* 290 (2023), <https://doi.org/10.1016/j.enbuild.2023.113106>.
- [62] Y. Zhou, H. Lei, X. Zhang, S. Wang, Y. Xu, C. Li, J. Zhang, Using the dual concept of evolutionary game and reinforcement learning in support of decision-making process of community regeneration—case study in Shanghai, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13010175>.
- [63] O. Hakimi, H. Liu, O. Abudayyeh, A. Houshyar, M. Almatared, A. Alhawiti, Data fusion for smart civil infrastructure management: a conceptual digital twin framework, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13112725>.
- [64] S. Kalantari, S. Pourjabar, T.B. Xu, J. Kan, Developing and user-testing a “digital twins” prototyping tool for architectural design, *Autom. Constr.* 135 (2022), <https://doi.org/10.1016/j.autcon.2022.104140>.
- [65] Z. Cheng, S. Tang, H. Liu, Z. Lei, Digital technologies in offsite and prefabricated construction: theories and applications, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13010163>.
- [66] T. Gao, Z. Yuanzhou, B. Ji, Z. Xie, Vision-based fatigue crack automatic perception and geometric updating of finite element model for welded joint in steel structures, *Comput. Aided Civ. Inf. Eng.* 39 (2024) 1659–1675, <https://doi.org/10.1111/mice.13166>.
- [67] M. Li, Q. Lu, S. Bai, M. Zhang, H. Tian, L. Qin, Digital twin-driven virtual sensor approach for safe construction operations of trailing suction hopper dredger, *Autom. Constr.* 132 (2021), <https://doi.org/10.1016/j.autcon.2021.103961>.
- [68] V.M. Gnecco, F. Vittori, A.L. Pisello, Digital twins for decoding human-building interaction in multi-domain test-rooms for environmental comfort and energy saving via graph representation, *Energ. Buildings* 279 (2023), <https://doi.org/10.1016/j.enbuild.2022.112652>.
- [69] G. Yu, D. Lin, Y. Wang, M. Hu, V. Sugumaran, J. Chen, Digital twin-enabled and knowledge-driven decision support for tunnel electromechanical equipment maintenance, *Tunn. Undergr. Space Technol.* 140 (2023), <https://doi.org/10.1016/j.tust.2023.105318>.
- [70] A. Hosseini Gourabpasi, M. Nik-Bakht, BIM-based automated fault detection and diagnostics of HVAC systems in commercial buildings, *J. Build. Eng.* 87 (2024), <https://doi.org/10.1016/j.jobte.2024.109022>.
- [71] M. Both, B. Kämper, A. Cartus, J. Beermann, T. Fessler, D.J. Müller, D.C. Diedrich, Automated monitoring applications for existing buildings through natural language processing based semantic mapping of operational data and creation of digital twins, *Energ. Buildings* 300 (2023), <https://doi.org/10.1016/j.enbuild.2023.113635>.
- [72] M. Mahmoud, W. Chen, Y. Yang, Y. Li, Automated BIM generation for large-scale indoor complex environments based on deep learning, *Autom. Constr.* 162 (2024), <https://doi.org/10.1016/j.autcon.2024.105376>.
- [73] L. Zhao, H. Zhang, Q. Wang, H. Wang, Digital-twin-based evaluation of nearly zero-energy building for existing buildings based on scan-to-BIM, *Adv. Civil Eng.* 2021 (2021), <https://doi.org/10.1155/2021/6638897>.
- [74] E. O'Dwyer, I. Pan, R. Charlesworth, S. Butler, N. Shah, Integration of an energy management tool and digital twin for coordination and control of multi-vector smart energy systems, *Sustain. Cities Soc.* 62 (2020), <https://doi.org/10.1016/j.scs.2020.102412>.
- [75] R. Ward, R. Choudary, M. Jans Singh, F. Roumpani, T. Lazauskas, M. Yong, N. Barlow, M. Hauru, The challenges of using live-streamed data in a predictive digital twin, *J. Build. Perform. Simul.* 16 (2023) 609–630, <https://doi.org/10.1080/19401493.2023.2187463>.
- [76] W. Xu, S. Liu, Novel economic models for advancing urban energy management and transition: simulation of urban energy system in digital twin, *Sustain. Cities Soc.* 101 (2024), <https://doi.org/10.1016/j.scs.2023.105154>.
- [77] Y. Zhao, N. Wang, Z. Liu, E. Mu, Construction theory for a building intelligent operation and maintenance system based on digital twins and machine learning, *Buildings* 12 (2022), <https://doi.org/10.3390/buildings12020087>.
- [78] J. Koo, S. Yoon, Neural network-based nonintrusive calibration for an unobserved model in digital twin-enabled building operations, *Autom. Constr.* 159 (2024), <https://doi.org/10.1016/j.autcon.2023.105261>.
- [79] Y. Wei, B. Akinci, Panorama-to-model registration through integration of image retrieval and semantic reprojection, *Autom. Constr.* 140 (2022), <https://doi.org/10.1016/j.autcon.2022.104356>.
- [80] H.H. Hosamo, H.K. Nielsen, D. Kraniotis, P.R. Svennevig, K. Svidt, Improving building occupant comfort through a digital twin approach: a Bayesian network model and predictive maintenance method, *Energ. Buildings* 288 (2023), <https://doi.org/10.1016/j.enbuild.2023.112992>.
- [81] T. Li, Y. Rui, S. Zhao, Y. Zhang, H. Zhu, X. Li, A quantitative digital twin maturity model for underground infrastructure based on D-ANP, *Tunn. Undergr. Space Technol.* 146 (2024), <https://doi.org/10.1016/j.tust.2024.105612>.
- [82] H. Gong, D. Su, S. Zeng, X. Chen, Parallel simulation and prediction techniques for digital twins in urban underground spaces, *Autom. Constr.* 175 (2025) 106212, <https://doi.org/10.1016/J.AUTCON.2025.106212>.
- [83] T.D.H.N. Nguyen, D.H. Ly, H. Jang, H.N.N. Dinh, Y. Ahn, Digital twin framework to enhance facility management for relocatable modular buildings, *Autom. Constr.* 176 (2025) 106249, <https://doi.org/10.1016/J.AUTCON.2025.106249>.
- [84] A.S. Borkowski, Digital twin conceptual framework for the operation and maintenance phase in the building's lifecycle, *Arch. Civ. Eng.* 70 (2024) 139–152, <https://doi.org/10.24425/ace.2024.150975>.
- [85] J. Schlenger, K. Pluta, A. Mathew, T. Yeung, R. Sacks, A. Borrmann, Reference architecture and ontology framework for digital twin construction, *Autom. Constr.* 174 (2025), <https://doi.org/10.1016/j.autcon.2025.106111>.
- [86] T. Jirawattanasomkul, L. Hang, S. Srivaranun, S. Likitlersuang, P. Jongvivatsakul, W. Yodsudjai, P. Thammarak, Digital twin-based structural health monitoring and measurements of dynamic characteristics in balanced cantilever bridge, *Resilient Cities Struct.* 4 (2025) 48–66, <https://doi.org/10.1016/J.RCNS.2025.08.001>.
- [87] S. Yoon, J. Lee, J. Li, P. Wang, Virtual in-situ modeling between digital twin and BIM for advanced building operations and maintenance, *Autom. Constr.* 168 (2024) 105823, <https://doi.org/10.1016/J.AUTCON.2024.105823>.
- [88] T. Li, Y. Rui, H. Zhu, L. Lu, X. Li, Comprehensive digital twin for infrastructure: a novel ontology and graph-based modelling paradigm, *Adv. Eng. Inform.* 62 (2024) 102747, <https://doi.org/10.1016/J.AEI.2024.102747>.
- [89] T. Ma, F. Xiao, C. Zhang, J. Zhang, H. Zhang, K. Xu, X. Luo, Digital twin for 3D interactive building operations: integrating BIM, IoT-enabled building automation systems, AI, and mixed reality, *Autom. Constr.* 176 (2025) 106277, <https://doi.org/10.1016/J.AUTCON.2025.106277>.
- [90] M. Afrazi, D. Jahed Armaghani, H. Afrazi, H. Fattahi, P. Samui, Real-time monitoring of tunnel structures using digital twin and artificial intelligence: a short overview, *Deep Undergr. Sci. Eng.* (2025), <https://doi.org/10.1002/DUG2.70029>.
- [91] H.H. Hosamo, P.R. Svennevig, K. Svidt, D. Han, H.K. Nielsen, A digital twin predictive maintenance framework of air handling units based on automatic fault detection and diagnostics, *Energ. Buildings* 261 (2022), <https://doi.org/10.1016/j.enbuild.2022.111988>.
- [92] S. Yu, D. Li, J. Ou, Digital twin-based structure health hybrid monitoring and fatigue evaluation of orthotropic steel deck in cable-stayed bridge, *Struct. Control. Health Monit.* 29 (2022), <https://doi.org/10.1002/stc.2976>.
- [93] A. Pal, J.J. Lin, S.H. Hsieh, M. Golparvar-Fard, Automated vision-based construction progress monitoring in built environment through digital twin, *Dev. Built Environ.* 16 (2023), <https://doi.org/10.1016/j.dibe.2023.100247>.
- [94] S. Li, Q. Yang, J. Xing, W. Chen, R. Zou, A foundation model for building digital twins: a case study of a chiller, *Buildings* 12 (2022), <https://doi.org/10.3390/buildings12081079>.
- [95] D. Sivori, L. Ierimonti, I. Venanzi, F. Ubertini, S. Cattari, An equivalent frame digital twin for the seismic monitoring of historic structures: a case study on the Consoli palace in Gubbio, Italy, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13071840>.
- [96] T.W. Kang, Y. Mo, A comprehensive digital twin framework for building environment monitoring with emphasis on real-time data connectivity and predictability, *Dev. Built Environ.* 17 (2024), <https://doi.org/10.1016/j.dibe.2023.100309>.
- [97] F. Jiang, L. Ma, T. Broyd, K. Chen, H. Luo, Underpass clearance checking in highway widening projects using digital twins, *Autom. Constr.* 141 (2022), <https://doi.org/10.1016/j.autcon.2022.104406>.
- [98] Z.Y. Wu, A. Chew, X. Meng, J. Cai, J. Pok, R. Kalfarisi, K.C. Lai, S.F. Hew, J. Wong, High fidelity digital twin-based anomaly detection and localization for smart water grid operation management, *Sustain. Cities Soc.* 91 (2023), <https://doi.org/10.1016/j.scs.2023.104446>.
- [99] Y. Pan, L. Zhang, A BIM-data mining integrated digital twin framework for advanced project management, *Autom. Constr.* 124 (2021), <https://doi.org/10.1016/j.autcon.2021.103564>.
- [100] K. Lin, Y.L. Xu, X. Lu, Z. Guan, J. Li, Digital twin-based collapse fragility assessment of a long-span cable-stayed bridge under strong earthquakes, *Autom. Constr.* 123 (2021), <https://doi.org/10.1016/j.autcon.2020.103547>.
- [101] W. Jiang, L. Ding, C. Zhou, Digital twin: stability analysis for tower crane hoisting safety with a scale model, *Autom. Constr.* 138 (2022), <https://doi.org/10.1016/j.autcon.2022.104257>.
- [102] S. Kookalani, D. Shen, Effect of fluid viscous damper parameters on the seismic performance, *J. Civil Eng. Mater. Appl.* 4 (2020) 141–153, <https://doi.org/10.12034/JCEMA.2020.232288.1025>.
- [103] X. Zhang, Y. Jiang, X. Wu, Z. Nan, Y. Jiang, J. Shi, Y. Zhang, X. Huang, G.G. Q. Huang, AIoT-enabled digital twin system for smart tunnel fire safety management, *Dev. Built Environ.* 18 (2024), <https://doi.org/10.1016/j.dibe.2024.100381>.
- [104] V. Hoskere, D. Hassanlou, A.U. Rahman, R. Bazrgary, M.T. Ali, Unified framework for digital twins of bridges, *Autom. Constr.* 175 (2025) 106214, <https://doi.org/10.1016/J.AUTCON.2025.106214>.

- [105] M.A. Talaghat, A. Golroo, A. Kharbouch, M. Rasti, R. Heikkilä, R. Jurva, Digital twin technology for road pavement, *Autom. Constr.* 168 (2024) 105826, <https://doi.org/10.1016/J.AUTCON.2024.105826>.
- [106] Y. Da, Y. Gao, Y. Li, D. Ren, K. Liu, A. Bras, A. Shaw, Advances in smart technologies and materials for automated asphalt pavement inspection: toward transport infrastructure digitalisation, *Autom. Constr.* 180 (2025) 106523, <https://doi.org/10.1016/J.AUTCON.2025.106523>.
- [107] G. Yu, S. Zhang, M. Hu, Y. Ken Wang, Prediction of highway tunnel pavement performance based on digital twin and multiple time series stacking, *Adv. Civil Eng.* 2020 (2020), <https://doi.org/10.1155/2020/8824135>.
- [108] L. Bellini Machado, M. Massao Futai, Tunnel performance prediction through degradation inspection and digital twin construction, *Tunn. Undergr. Space Technol.* 144 (2024), <https://doi.org/10.1016/j.tust.2023.105544>.
- [109] J. Li, Q. Li, Q. Mao, H. Xu, Digital-twin-based high-precision assembly of a steel bridge tower, *Buildings* 13 (2023) 257, <https://doi.org/10.3390/buildings13010257>.
- [110] X. Hu, R.H. Assaad, A BIM-enabled digital twin framework for real-time indoor environment monitoring and visualization by integrating autonomous robotics, LiDAR-based 3D mobile mapping, IoT sensing, and indoor positioning technologies, *J. Build. Eng.* 86 (2024), <https://doi.org/10.1016/j.jobe.2024.108901>.
- [111] Z. Liu, W. Bai, X. Du, A. Zhang, Z. Xing, A. Jiang, Digital twin-based safety evaluation of prestressed steel structure, *Adv. Civil Eng.* 2020 (2020), <https://doi.org/10.1155/2020/8888876>.
- [112] J. Heng, L. Lai, Y. Dong, S. Kaewunruen, C. Baniotopoulos, Digital twin for intelligent maintenance towards sustainable bridges, bridge maintenance, safety, management, digitalization and sustainability, in: *Proceedings of the 12th International Conference on Bridge Maintenance, Safety and Management 2024, IABMAS, 2024*, pp. 693–700, <https://doi.org/10.1201/9781003483755-80/DIGITAL-TWIN-INTELLIGENT-MAINTENANCE-TOWARDS-SUSTAINABLE-BRIDGES-HENG-LAI-DONG-KAEWUNRUEN-BANIOTOPOULOS>.
- [113] K. Lin, Y.L. Xu, X. Lu, Z. Guan, J. Li, Digital twin-based life-cycle seismic performance assessment of a long-span cable-stayed bridge, *Bull. Earthq. Eng.* 21 (2023) 1203–1227, <https://doi.org/10.1007/S10518-022-01567-W/TABLES/4>.
- [114] H.H. Hosamo, M.H. Hosamo, Digital twin technology for bridge maintenance using 3D laser scanning: a review, *Adv. Civil Eng.* 2022 (2022), <https://doi.org/10.1155/2022/2194949>.
- [115] A. Consilvio, J.S. Hernández, W. Chen, I. Brilakis, L. Bartoccini, F. Di Gennaro, M. van Welie, Towards a digital twin-based intelligent decision support for road maintenance, *Transp. Res. Procedia* 69 (2023) 791–798, <https://doi.org/10.1016/j.trpro.2023.02.237>.
- [116] F. Jiang, L. Ma, T. Broyd, W. Chen, H. Luo, Digital twin enabled sustainable urban road planning, *Sustain. Cities Soc.* 78 (2022), <https://doi.org/10.1016/j.scs.2021.103645>.
- [117] F. Zahedi, H. Alavi, J. Majrouhi Sardroud, H. Dang, Digital twins in the sustainable construction industry, *Buildings* 14 (2024) 3613, <https://doi.org/10.3390/buildings14113613>.
- [118] R. Bortolini, R. Rodrigues, H. Alavi, L.F.D. Vecchia, N. Forcada, Digital twins' applications for building energy efficiency: a review, *Energies* 15 (2022) 7002, <https://doi.org/10.3390/EN15197002>.
- [119] H. Alavi, S. Kookalani, F. Rahimian, N. Forcada, *Integr. Build. Intellig.* (2024), <https://doi.org/10.1007/978-3-031-68865-2>.
- [120] H. Alavi, S. Kookalani, F. Rahimian, N. Forcada, BIM-based DSS for enhancing occupants' comfort, *Integr. Build. Intellig.* (2024) 79–99, https://doi.org/10.1007/978-3-031-68865-2_6.
- [121] S.L. Fan, W.S. Ong, C.T. Wu, N. Forcada Matheu, H. Alavi, Augmented reality-based facility maintenance management system, *Facilities* 41 (2023) 769–800, <https://doi.org/10.1108/F-04-2022-0059/FULL/XML>.
- [122] H. Alavi, P. Gordo-Gregorio, N. Forcada, A. Bayramova, D.J. Edwards, AI-driven BIM integration for optimizing healthcare facility design, *Buildings* 14 (2024), <https://doi.org/10.3390/buildings14082354>.
- [123] H. Alavi, S. Kookalani, F. Rahimian, N. Forcada, BIM-based augmented reality for facility maintenance management, *Integr. Build. Intellig.* (2024) 101–112, https://doi.org/10.1007/978-3-031-68865-2_7.
- [124] P. Gordo-Gregorio, H. Alavi, D.J. Edwards, N. Forcada, F. Guéna, An occupant-centric approach on digital twins for building management, *Build. Res. Inf.* (2025), <https://doi.org/10.1080/09613218.2025.2463676;PAGE=STRING:ARTICLE/CHAPTER>.
- [125] L. Lu, A.M. d'Avigneau, Y. Pan, Z. Sun, P. Luo, I. Brilakis, Modeling heterogeneous spatiotemporal pavement data for condition prediction and preventive maintenance in digital twin-enabled highway management, *Autom. Constr.* 174 (2025) 106134, <https://doi.org/10.1016/J.AUTCON.2025.106134>.
- [126] D. Gürdür Broo, M. Bravo-Haro, J. Schooling, Design and implementation of a smart infrastructure digital twin, *Autom. Constr.* 136 (2022), <https://doi.org/10.1016/j.autcon.2022.104171>.
- [127] Y. Ding, Y. Zhang, X. Huang, Intelligent emergency digital twin system for monitoring building fire evacuation, *J. Build. Eng.* 77 (2023), <https://doi.org/10.1016/j.jobe.2023.107416>.
- [128] M. Kamari, Y. Ham, AI-based risk assessment for construction site disaster preparedness through deep learning-based digital twinning, *Autom. Constr.* 134 (2022), <https://doi.org/10.1016/j.autcon.2021.104091>.
- [129] H.H. Hosamo, A. Imran, J. Cardenas-Cartagena, P.R. Svennevig, K. Svidt, H. K. Nielsen, A review of the digital twin technology in the AEC-FM industry, *Adv. Civil Eng.* 2022 (2022), <https://doi.org/10.1155/2022/2185170>.
- [130] D.G.J. Opoku, S. Perera, R. Osei-Kyei, M. Rashidi, T. Famakinwa, K. Bamdad, Drivers for digital twin adoption in the construction industry: a systematic literature review, *Buildings* 12 (2022), <https://doi.org/10.3390/buildings12020113>.
- [131] S. Siccardi, V. Villa, Trends in adopting BIM, IoT and DT for facility management: a scientometric analysis and keyword co-occurrence network review, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13010015>.
- [132] Z. Ni, C. Zhang, M. Karlsson, S. Gong, A study of deep learning-based multi-horizon building energy forecasting, *Energ. Buildings* 303 (2024), <https://doi.org/10.1016/j.enbuild.2023.113810>.
- [133] X. Li, J. Luo, Y. Li, W. Wang, W. Hong, M. Liu, X. Li, Z. Lv, Application of effective water-energy management based on digital twins technology in sustainable cities construction, *Sustain. Cities Soc.* 87 (2022), <https://doi.org/10.1016/j.scs.2022.104241>.
- [134] H. Yue, Q. Wang, M. Zhao, Z. Yang, L. Lu, Advancements in digital twin applications for intelligent construction quality management, *J. Constr. Eng. Manag.* 152 (2025), <https://doi.org/10.1061/JCEMD4.COENG-17126>.
- [135] Z. Yang, C. Tang, T. Zhang, Z. Zhang, D.T. Doan, Digital twins in construction: architecture, applications, trends and challenges, *Buildings* 14 (2024), <https://doi.org/10.3390/buildings14092616>.
- [136] G. Piras, F. Muzi, V.A. Tiburcio, Digital management methodology for building production optimization through digital twin and artificial intelligence integration, *Buildings* 14 (2024), <https://doi.org/10.3390/buildings14072110>.
- [137] A. Costin, A. Adibfar, J. Bridge, Digital twin framework for bridge structural health monitoring utilizing existing technologies: new paradigm for enhanced management, operation, and maintenance, *Transp. Res. Rec.* 2678 (2024) 1095–1106, <https://doi.org/10.1177/03611981231208908>.
- [138] P. Du, X. Zhao, Z. Zhang, X. Sun, G. Du, Digital twin-based numerical simulation method for Cee-shape cold-formed steel members, *Buildings* 13 (2023), <https://doi.org/10.3390/buildings13092388>.
- [139] L. Zhang, J. Guo, X. Fu, R.L.K. Tiong, P. Zhang, Digital twin enabled real-time advanced control of TBM operation using deep learning methods, *Autom. Constr.* 158 (2024), <https://doi.org/10.1016/j.autcon.2023.105240>.
- [140] Y. Zhao, C. Cao, Z. Liu, A framework for prefabricated component hoisting management systems based on digital twin technology, *Buildings* 12 (2022), <https://doi.org/10.3390/buildings12030276>.
- [141] Z. Liu, J. Wang, Protection and utilization of historical sites using digital twins, *Buildings* 14 (2024), <https://doi.org/10.3390/buildings14041019>.
- [142] P. Luo, S.A. Prieto, E. Parn, B. García de Soto, I. Brilakis, LLM-driven Human-robot Interaction with Digital Twins for Facility Management, 2025, <https://doi.org/10.22260/CCC2025/0061>.
- [143] P. Luo, E. Parn, I. Brilakis, ChatTwin: enabling natural language interactions with infrastructure digital twins, in: *Proceedings of the European Conference on Computing in Construction 5, 2024*, <https://doi.org/10.35490/EC3.2024.259>.
- [144] P. Luo, X. Wen, Y. Jiang, E. Parn, I. Brilakis, ChatTwin: bridging the usability gap to digital twin adoption in infrastructure operations and maintenance with a natural language interface, *Autom. Constr.* 181 (2026) 106655, <https://doi.org/10.1016/J.AUTCON.2025.106655>.
- [145] L. Lu, Y. Pan, B. Sheil, P. Luo, I. Brilakis, Regional graph-based pavement condition data quality assessment in support of trustworthy highway infrastructure digital twins, *Autom. Constr.* 181 (2026) 106603, <https://doi.org/10.1016/J.AUTCON.2025.106603>.
- [146] F. Mok, P. Luo, W. Chen, I. Brilakis, Enriching and maintaining road digital twins with condition information, in: *The 30th EG-ICE: International Conference on Intelligent Computing in Engineering, 2023*. <https://www.repository.cam.ac.uk/items/57545721-73f7-4820-a337-1a6d1a6d9457>.